

islr-ch2-ch3

Rohit Goswami

```
+++ title = "ISLR :: Multiple Linear Regression" author = ["Rohit Goswami"] date = 2020-01-15T05:28:00+00:00 tags = ["solutions", "R", "ISLR"] categories = ["programming"] draft = false toc = true comments = true +++
```

Chapter II - Statistical Learning

All the questions are as per the ISL seventh printing of the First edition [^fn:1].

Question 2.8 - Pages 54-55

This exercise relates to the `College` data set, which can be found in the file `College.csv`. It contains a number of variables for 777 different universities and colleges in the US. The variables are

- `Private` : Public/private indicator
- `Apps` : Number of applications received
- `Accept` : Number of applicants accepted
- `Enroll` : Number of new students enrolled
- `Top10perc` : New students from top 10 % of high school class
- `Top25perc` : New students from top 25 % of high school class
- `F.Undergrad` : Number of full-time undergraduates
- `P.Undergrad` : Number of part-time undergraduates
- `Outstate` : Out-of-state tuition
- `Room.Board` : Room and board costs
- `Books` : Estimated book costs
- `Personal` : Estimated personal spending
- `PhD` : Percent of faculty with Ph.D.'s
- `Terminal` : Percent of faculty with terminal degree
- `S.F.Ratio` : Student/faculty ratio
- `perc.alumni` : Percent of alumni who donate
- `Expend` : Instructional expenditure per student
- `Grad.Rate` : Graduation rate

Before reading the data into R, it can be viewed in Excel or a text editor.

(a) Use the `read.csv()` function to read the data into R. Call the loaded data `college`. Make sure that you have the directory set to the correct location for the data.

(b) Look at the data using the `fix()` function. You should notice that the first column is just the name of each university. We don't really want R to treat this as data. However, it may be handy to have these names for later. Try the following commands:

```
rownames(college)=college[,1]
fix(college)
```

You should see that there is now a row.names column with the name of each university recorded. This means that R has given each row a name corresponding to the appropriate university. R will not try to perform calculations on the row names. However, we still need to eliminate the first column in the data where the names are stored. Try:

```
college=college[,-1]
fix(college)
```

(c)

1. Use the `summary()` function to produce a numerical summary of the variables in the data set.
2. Use the `pairs()` function to produce a scatterplot matrix of the first ten columns or variables of the data. Recall that you can reference the first ten columns of a matrix `A` using `A[,1:10]`.
3. Use the `plot()` function to produce side-by-side boxplots of Outstate versus Private.
4. Create a new qualitative variable, called Elite, by binning the `Top10perc` variable. We are going to divide universities into two groups based on whether or not the proportion of students coming from the top 10 of their high school classes exceeds 50.

```
Elite = rep("No", nrow(college))
Elite [college$Top10perc >50]="Yes"
Elite = as.factor (Elite)
college = data.frame (college, Elite)
```

Use the `summary()` function to see how many elite universities there are. Now use the `plot()` function to produce side-by-side boxplots of Outstate versus Elite.

1. Use the `hist()` function to produce some histograms with differing numbers of bins for a few of the quantitative variables. You may find the command `par(mfrow=c(2,2))` useful: it will divide the print window into four regions so that four plots can be made simultaneously. Modifying the arguments to this function will divide the screen in other ways.
2. Continue exploring the data, and provide a brief summary of what you discover.

Answer

Instead of reading in data, for ISLR in particular we can load the [ISLR library](#) which is on CRAN and contains the data-sets required for the book.

```
install.packages("ISLR")
```

Thus, we can now read it in as `library("ISLR")`

The remaining sections are meant to be executed, and are marked as such, with `r` in `{}`.

(c)

We will load the dataset once for the whole document.

```
library("ISLR")
```

1. Usage of the `summary()` function

```
summary(ISLR::College)
```

```
## Private      Apps      Accept      Enroll      Top10perc
## No :212  Min.   :   81  Min.    :   72  Min.    :   35  Min.    : 1.00
## Yes:565  1st Qu.:  776  1st Qu.:  604  1st Qu.:  242  1st Qu.:15.00
##      Median : 1558  Median : 1110  Median :  434  Median :23.00
##      Mean   : 3002  Mean   : 2019  Mean   :  780  Mean   :27.56
##      3rd Qu.: 3624  3rd Qu.: 2424  3rd Qu.:  902  3rd Qu.:35.00
##      Max.   :48094  Max.   :26330  Max.   :6392  Max.   :96.00
## Top25perc    F.Undergrad    P.Undergrad    Outstate
## Min.   :   9.0  Min.    :  139  Min.    :    1.0  Min.    : 2340
## 1st Qu.:  41.0  1st Qu.:  992  1st Qu.:   95.0  1st Qu.: 7320
## Median :  54.0  Median : 1707  Median :  353.0  Median : 9990
## Mean   :  55.8  Mean   : 3700  Mean   :  855.3  Mean   :10441
## 3rd Qu.:  69.0  3rd Qu.: 4005  3rd Qu.:  967.0  3rd Qu.:12925
## Max.   :100.0  Max.   :31643  Max.   :21836.0  Max.   :21700
## Room.Board    Books      Personal      PhD
## Min.   :1780  Min.    :   96.0  Min.    :  250  Min.    :   8.00
## 1st Qu.:3597  1st Qu.:  470.0  1st Qu.:  850  1st Qu.:  62.00
## Median :4200  Median :  500.0  Median :1200  Median :  75.00
## Mean   :4358  Mean   :  549.4  Mean   :1341  Mean   :  72.66
## 3rd Qu.:5050  3rd Qu.:  600.0  3rd Qu.:1700  3rd Qu.:  85.00
## Max.   :8124  Max.   :2340.0  Max.   :6800  Max.   :103.00
## Terminal      S.F.Ratio      perc.alumni      Expend
## Min.   :  24.0  Min.    :   2.50  Min.    :   0.00  Min.    :  3186
## 1st Qu.:  71.0  1st Qu.:11.50  1st Qu.:13.00  1st Qu.: 6751
## Median :  82.0  Median :13.60  Median :21.00  Median : 8377
## Mean   :  79.7  Mean   :14.09  Mean   :22.74  Mean   : 9660
## 3rd Qu.:  92.0  3rd Qu.:16.50  3rd Qu.:31.00  3rd Qu.:10830
## Max.   :100.0  Max.   :39.80  Max.   :64.00  Max.   :56233
## Grad.Rate
## Min.   : 10.00
## 1st Qu.: 53.00
## Median : 65.00
## Mean   : 65.46
## 3rd Qu.: 78.00
## Max.   :118.00
```

1. Usage of `pairs()`

```
tenColl <- ISLR::College[,1:10] # For getting the first ten columns
pairs(tenColl) # Scatterplot
```

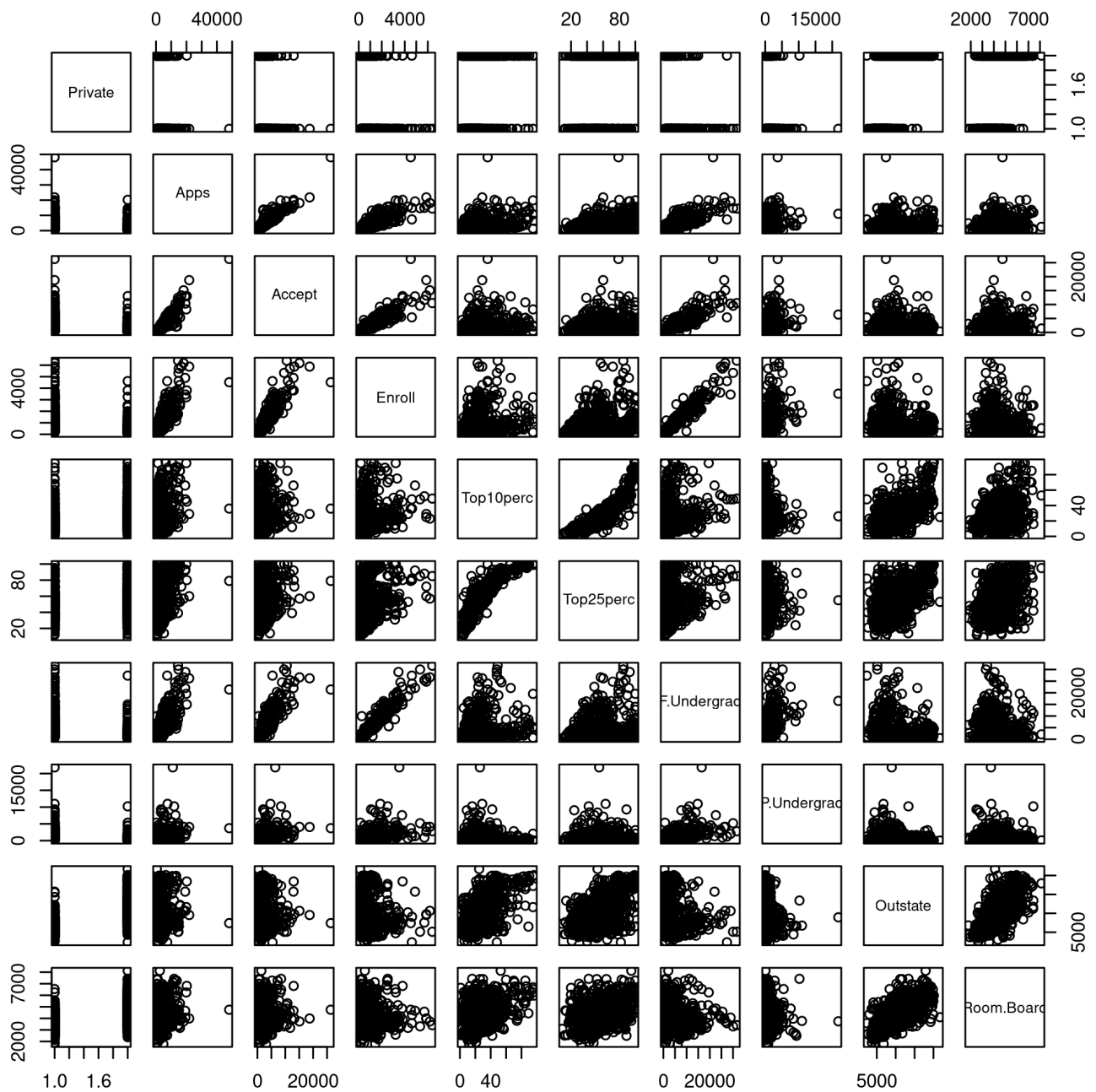


Figure 1: Pairs

1. Boxplot creation with `plot()`

```
plot(ISLR::College$Private, ISLR::College$Outstate, xlab="Private", ylab="Outstate")
```

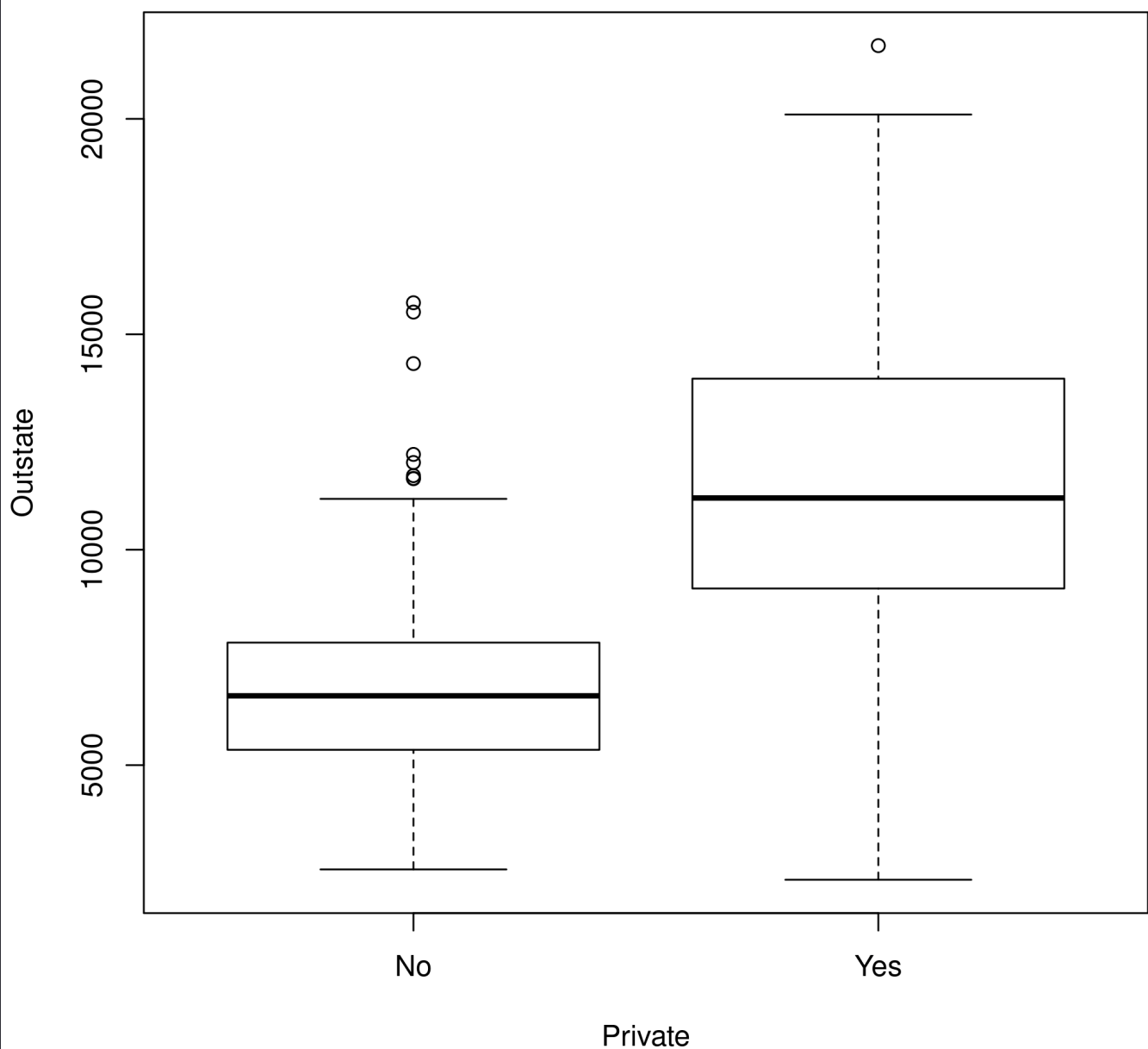


Figure 2: Boxplots

1. Binning and plotting

```
college=ISLR::College
Elite=rep("No",nrow(college))
Elite[college$Top10perc>50]="Yes"
Elite=as.factor(Elite)
college<-data.frame(college,Elite)
summary(college$Elite)
```

```
## No Yes
## 699 78
```

```
plot(college$Outstate,college$Elite,xlab="Outstate",ylab="Elite")
```

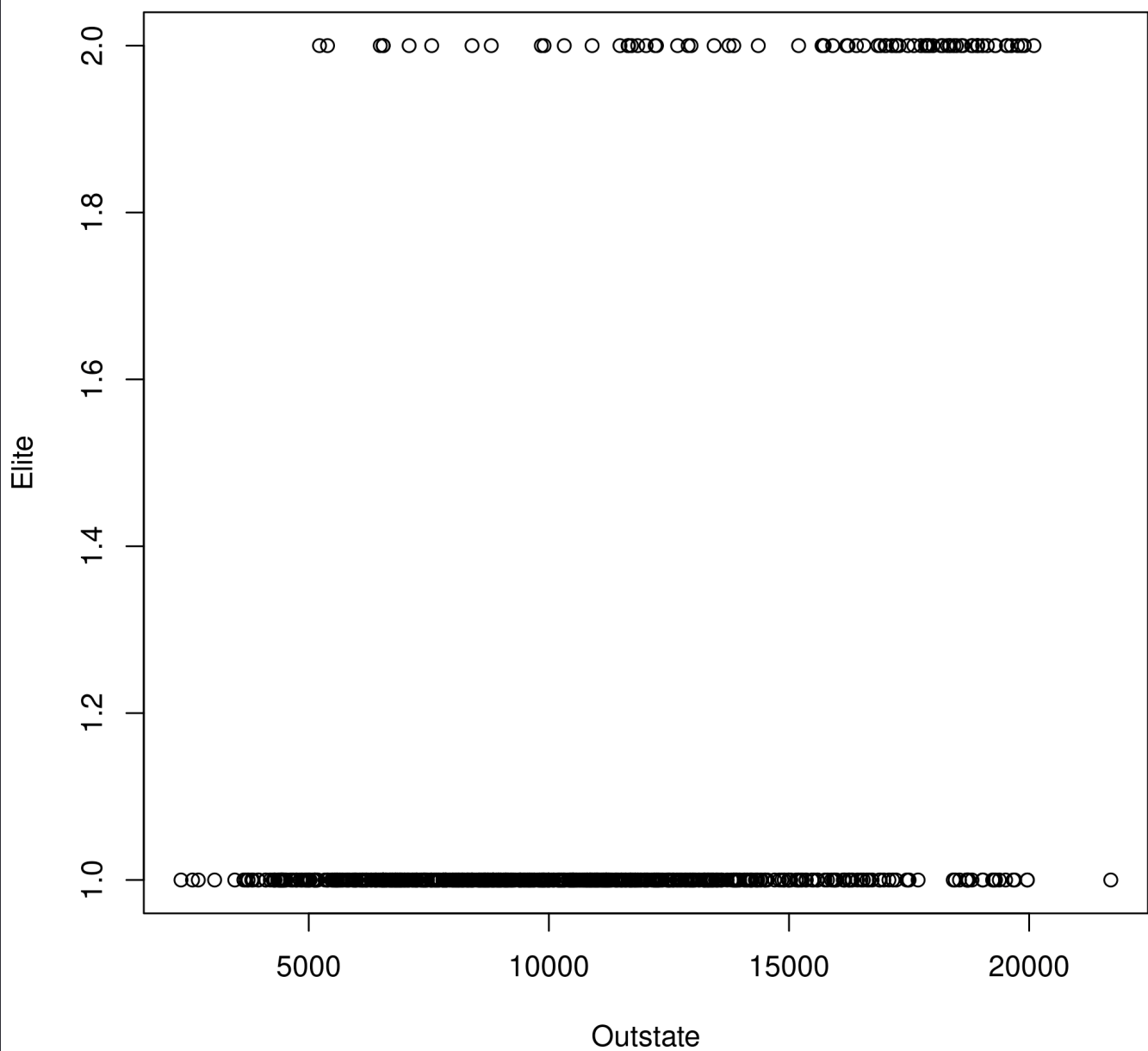


Figure 3: Plotting Outstate and Elite

1. Histograms with `hist()`

```
par(mfrow=c(2,2))
hist(college$Enroll)
hist(college$perc.alumni, col=2)
hist(college$Personal, col=3, breaks=10)
hist(college$PhD, breaks=10)
```

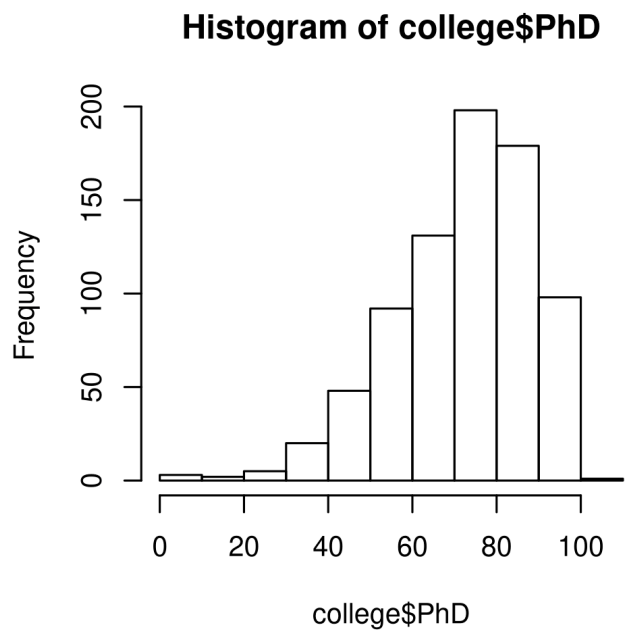
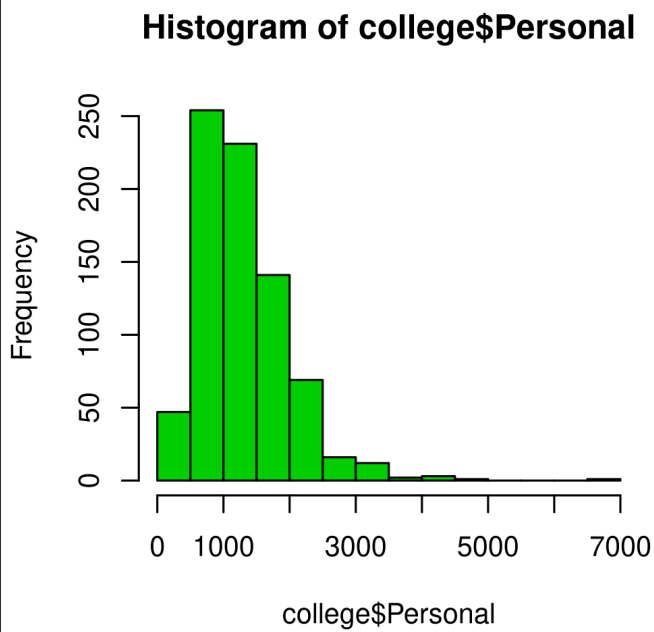
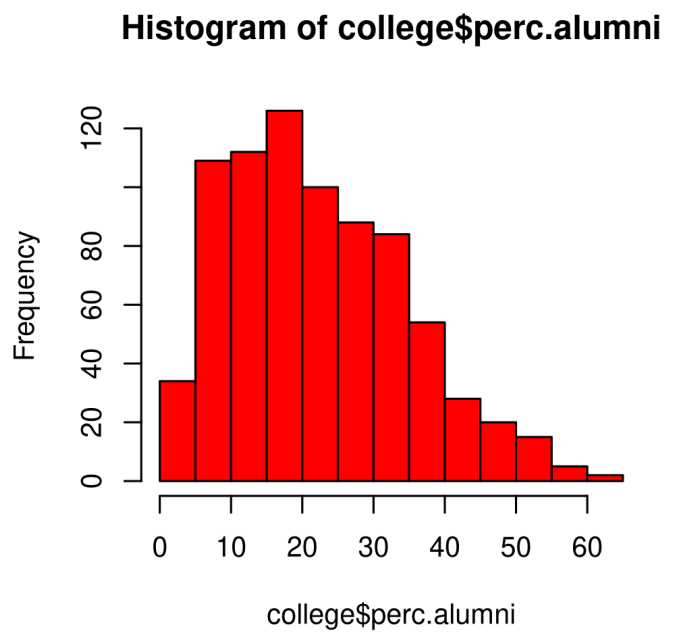
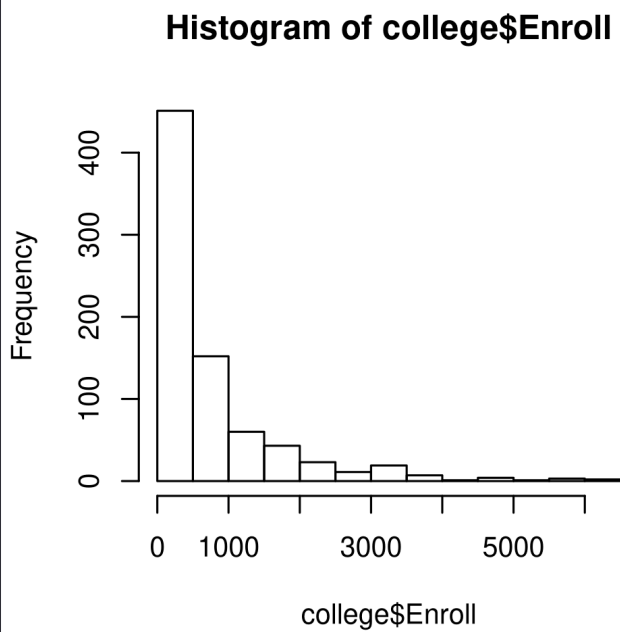


Figure 4: Histogram

```
hist(college$Top10perc, col="blue")  
hist(college$Outstate, col=23)
```

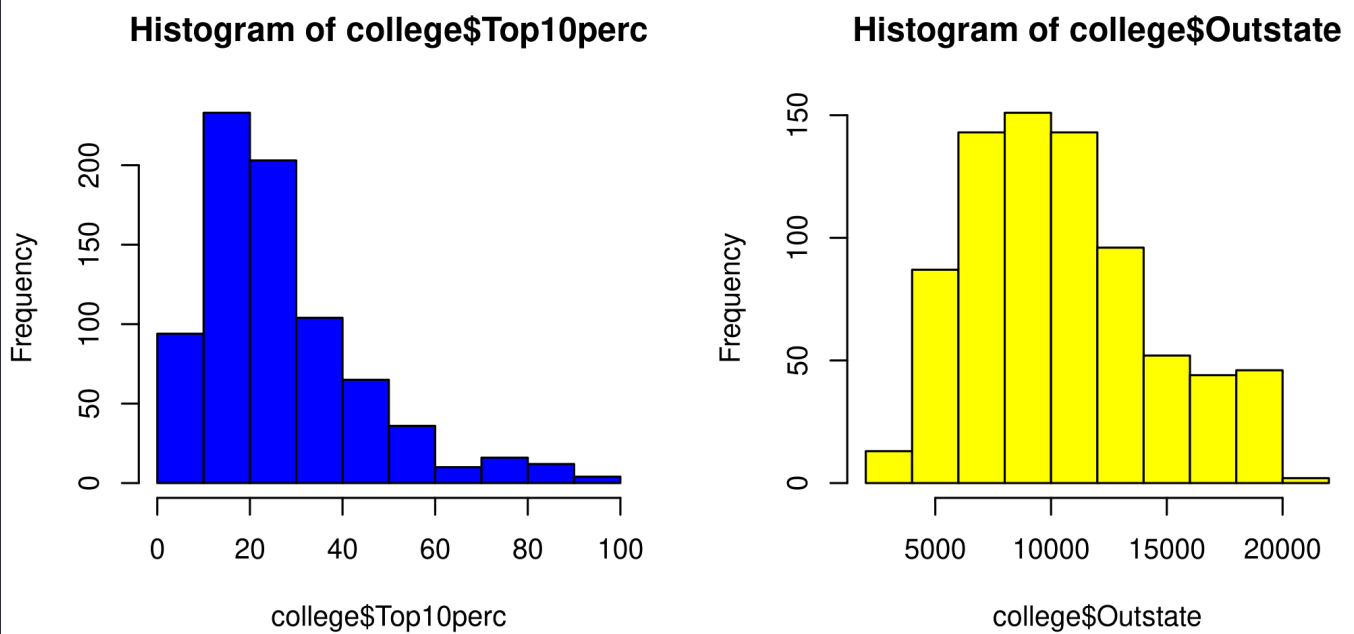


Figure 5: Colored Histogram

1. Explorations (graphical)

0 implies the faculty have PhDs. It is clear that people donate more when faculty do not have terminal degrees.

```
plot(college$Terminal-college$PhD, college$perc.alumni)
```

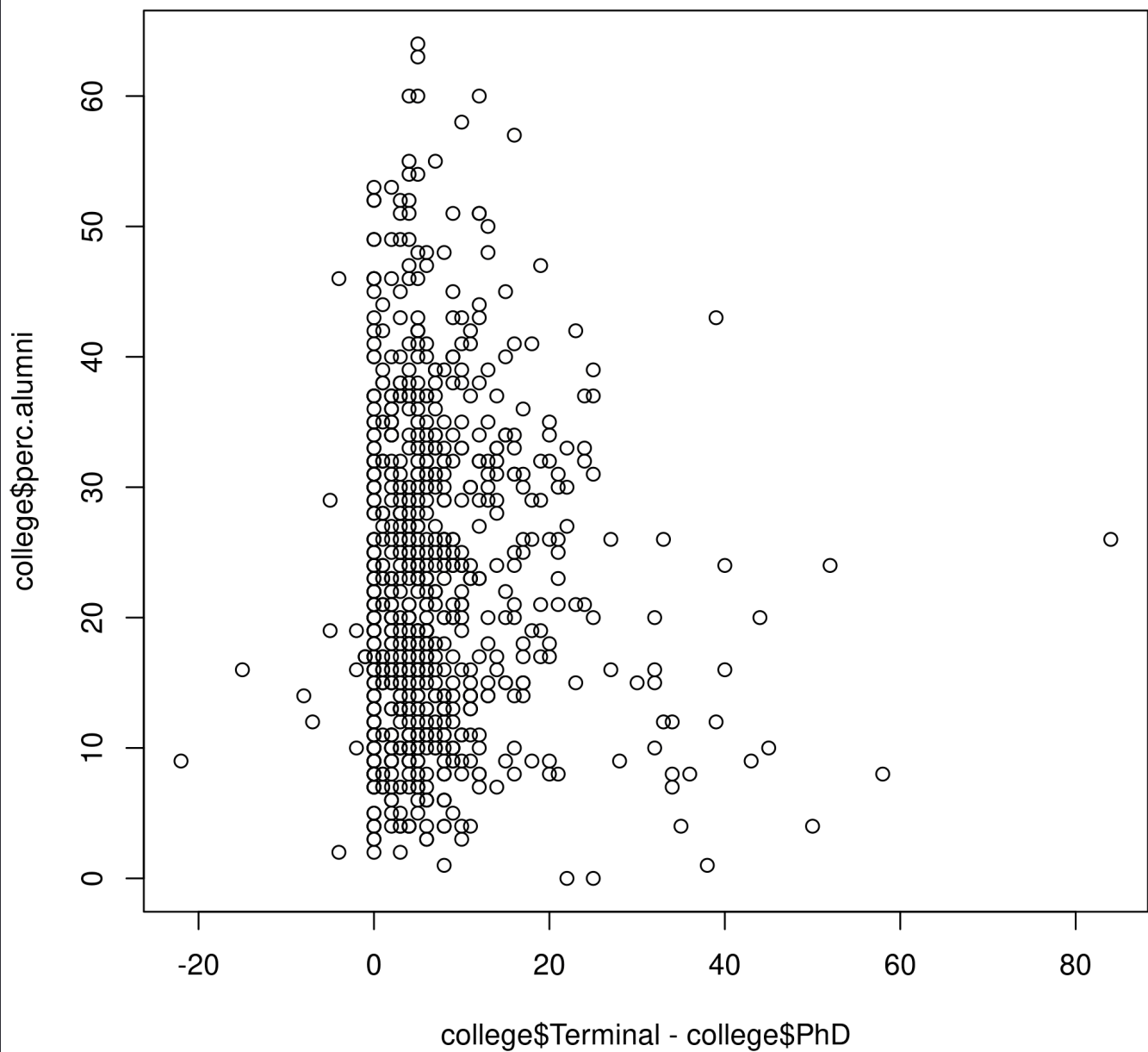



Figure 6: Terminal degrees and alumni

High tuition correlates to high graduation rate.

```
plot(college$Expend, college$Grad.Rate)
```

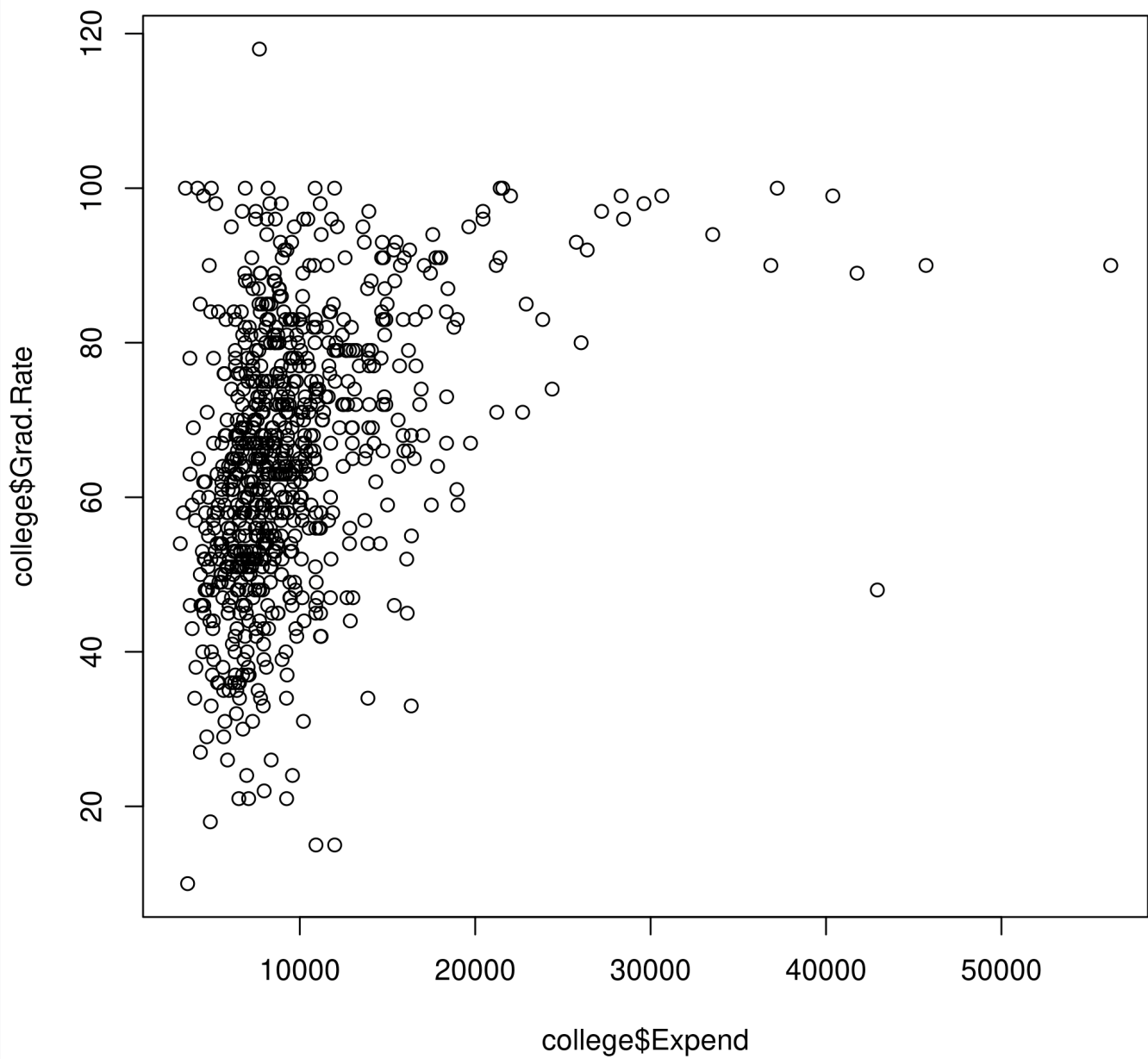


Figure 7: Tuiton and graduation

Low acceptance implies a low student to faculty ratio.

```
plot(college$Accept / college$Apps, college$S.F.Ratio)
```

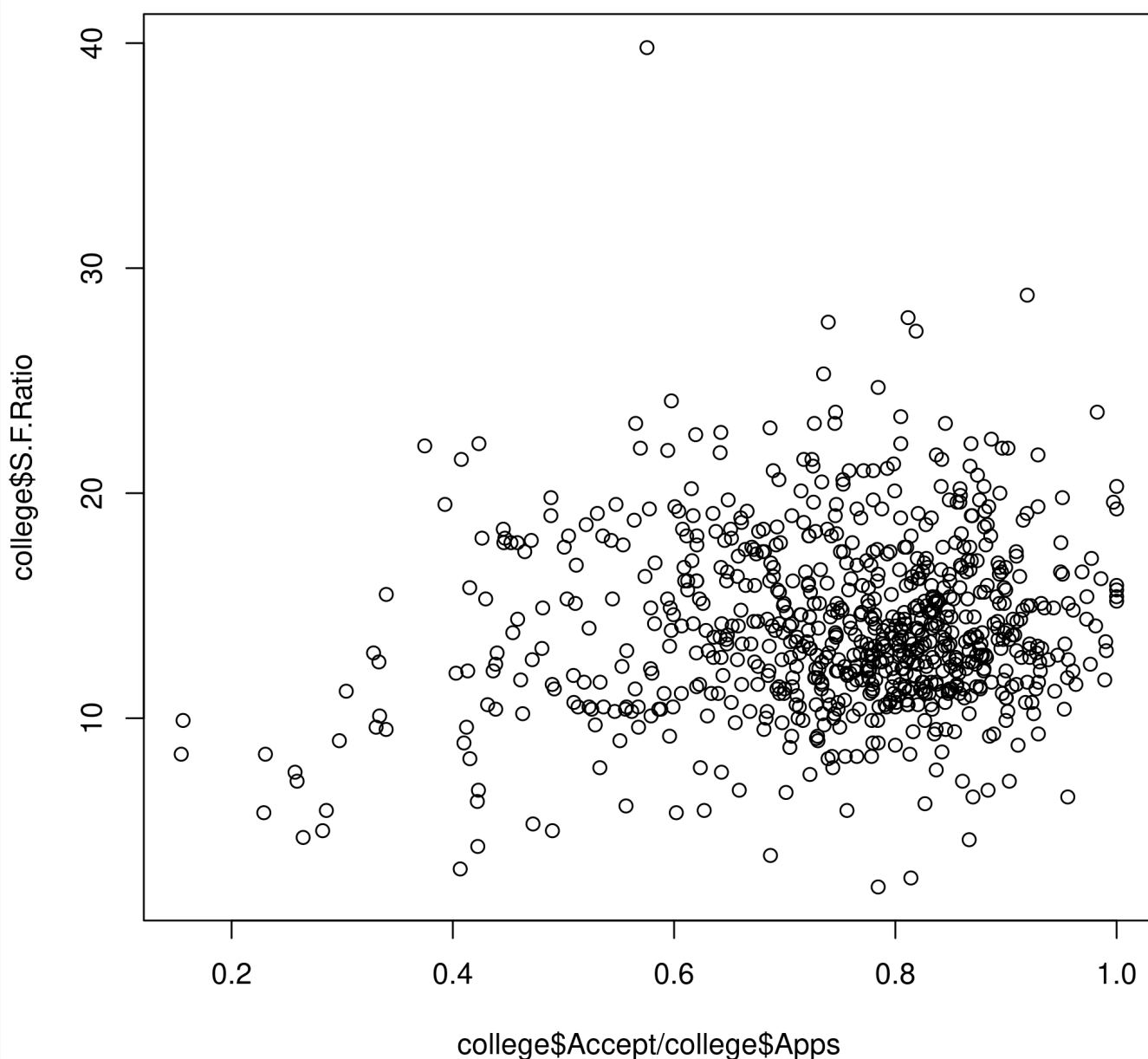


Figure 8: Acceptance and Student/Faculty ratio

Question 2.9 - Page 56

This exercise involves the `Auto` data set studied in the lab. Make sure that the missing values have been removed from the data.

- (a) Which of the predictors are quantitative, and which are qualitative?
- (b) What is the *range* of each quantitative predictor? You can answer this using the `range()` function.
- (c) What is the mean and standard deviation of each quantitative predictor?
- (d) Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?
- (e) Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.
- (f) Suppose that we wish to predict gas mileage (`mpg`) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting `mpg`? Justify your answer.

Answer

Once again, since the dataset is loaded from the library, we will simply start manipulating it.

```
# Clean data
autoDat<-na.omit(ISLR::Auto) # renamed for convenience
```

(a) To determine whether the variables are qualitative or quantitative we can either inspect the variables by eye, or query the dataset.

```
summary(autoDat) # Observe the output for variance
```

```
##      mpg      cylinders  displacement  horsepower      weight
##  Min.   : 9.00    Min.   :3.000    Min.   : 68.0    Min.   : 46.0    Min.   :1613
##  1st Qu.:17.00    1st Qu.:4.000    1st Qu.:105.0    1st Qu.: 75.0    1st Qu.:2225
##  Median :22.75    Median :4.000    Median :151.0    Median : 93.5    Median :2804
##  Mean   :23.45    Mean   :5.472    Mean   :194.4    Mean   :104.5    Mean   :2978
##  3rd Qu.:29.00    3rd Qu.:8.000    3rd Qu.:275.8    3rd Qu.:126.0    3rd Qu.:3615
##  Max.   :46.60    Max.   :8.000    Max.   :455.0    Max.   :230.0    Max.   :5140
##
##  acceleration      year      origin      name
##  Min.   : 8.00    Min.   :70.00    Min.   :1.000    amc matador      : 5
##  1st Qu.:13.78    1st Qu.:73.00    1st Qu.:1.000    ford pinto       : 5
##  Median :15.50    Median :76.00    Median :1.000    toyota corolla   : 5
##  Mean   :15.54    Mean   :75.98    Mean   :1.577    amc gremlin      : 4
##  3rd Qu.:17.02    3rd Qu.:79.00    3rd Qu.:2.000    amc hornet       : 4
##  Max.   :24.80    Max.   :82.00    Max.   :3.000    chevrolet chevette: 4
##                                     (Other)      :365
```

```
str(autoDat) # Directly find out
```

```
## 'data.frame': 392 obs. of 9 variables:
## $ mpg : num 18 15 18 16 17 15 14 14 14 15 ...
## $ cylinders : num 8 8 8 8 8 8 8 8 8 8 ...
## $ displacement: num 307 350 318 304 302 429 454 440 455 390 ...
## $ horsepower : num 130 165 150 150 140 198 220 215 225 190 ...
## $ weight : num 3504 3693 3436 3433 3449 ...
## $ acceleration: num 12 11.5 11 12 10.5 10 9 8.5 10 8.5 ...
## $ year : num 70 70 70 70 70 70 70 70 70 70 ...
## $ origin : num 1 1 1 1 1 1 1 1 1 1 ...
## $ name : Factor w/ 304 levels "amc ambassador brougham",...: 49 36 231 14 161 141 54 223 241 2 ...
```

From the above view, we can see that there is only one listed as a qualitative variable or factor, and that is **name**. However, we can also do this in a cleaner manner or at-least in a different manner with a function.

```
findFactors <- sapply(autoDat,is.factor)
findFactors
```

```
##      mpg      cylinders  displacement  horsepower      weight acceleration
##      FALSE      FALSE      FALSE      FALSE      FALSE      FALSE
##      year      origin      name
##      FALSE      FALSE      TRUE
```

Though only **name** is listed as a qualitative variable, we note that **origin** seems to be almost qualitative as well.

```
length(unique(autoDat$origin))
```

```
## [1] 3
```

```
unique(autoDat$origin)
```

```
## [1] 1 3 2
```

Infact we can check that nothing else has this property by repeated application of `sapply`, though a pipe would be more satisfying

```
getUniq<-sapply(autoDat, unique)
getLengths<-sapply(getUniq,length)
getLengths
```

```
##      mpg  cylinders displacement  horsepower      weight acceleration
##      127         5          81         93        346         95
##      year      origin         name
##      13         3         301
```

This is really nicer with pipes

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
## filter, lag
```

```
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
autoDat %>% sapply(unique) %>% sapply(length)
```

```
##      mpg  cylinders displacement  horsepower      weight acceleration
##      127         5          81         93        346         95
##      year      origin         name
##      13         3         301
```

At any rate, we know now that **origin** and **name** are probably qualitative, and the rest are quantitative.

(b) Using `range()`

A nice feature of the dataset we have is that the suspected qualitative variables are at the end of the dataset. So we can simply select the first 7 rows and go nuts on them.

```
autoDat[,1:7] %>% sapply(range) # or sapply(autoDat[,1:7],range)
```

```
##      mpg cylinders displacement horsepower weight acceleration year
## [1,]  9.0         3           68         46  1613           8.0   70
## [2,] 46.6         8          455        230  5140          24.8   82
```

Once again, more elegant with pipes and `subset()`

```
autoDat %>% subset(select=-c(name,origin)) %>% sapply(range)
```

```
##      mpg cylinders displacement horsepower weight acceleration year
## [1,]  9.0         3           68         46  1613           8.0   70
## [2,] 46.6         8          455        230  5140          24.8   82
```

```
# Even simpler with dplyr
autoDat %>% select(-name,-origin) %>% sapply(range)
```

```
##      mpg cylinders displacement horsepower weight acceleration year
## [1,]  9.0         3           68         46  1613           8.0   70
## [2,] 46.6         8          455        230  5140          24.8   82
```

(c) Mean and standard deviation

```
noFactors <- autoDat %>% select(-name,-origin)
noFactors %>% sapply(mean)
```

```
##      mpg      cylinders displacement      horsepower      weight acceleration
## 23.445918  5.471939  194.411990  104.469388  2977.584184  15.541327
##      year
## 75.979592
```

```
noFactors %>% sapply(sd)
```

```
##      mpg      cylinders displacement      horsepower      weight acceleration
##  7.805007  1.705783  104.644004   38.491160  849.402560   2.758864
##      year
##  3.683737
```

(d) Removing observations 10-85 and testing.

```
noFactors[-(10:85),] %>% sapply(mean)
```

```
##      mpg      cylinders displacement      horsepower      weight acceleration
## 24.404430  5.373418  187.240506  100.721519  2935.971519  15.726899
##      year
## 77.145570
```

```
noFactors[-(10:85),] %>% sapply(sd)
```

```
##      mpg      cylinders displacement      horsepower      weight acceleration
##  7.867283  1.654179   99.678367   35.708853  811.300208   2.693721
##      year
##  3.106217
```

(e) Plots for determining relationships

```
par(mfrow=c(2,2))
plot(autoDat$weight, autoDat$horsepower)
plot(autoDat$weight, autoDat$acceleration)
plot(autoDat$displacement, autoDat$acceleration)
plot(autoDat$cylinders, autoDat$acceleration)
```

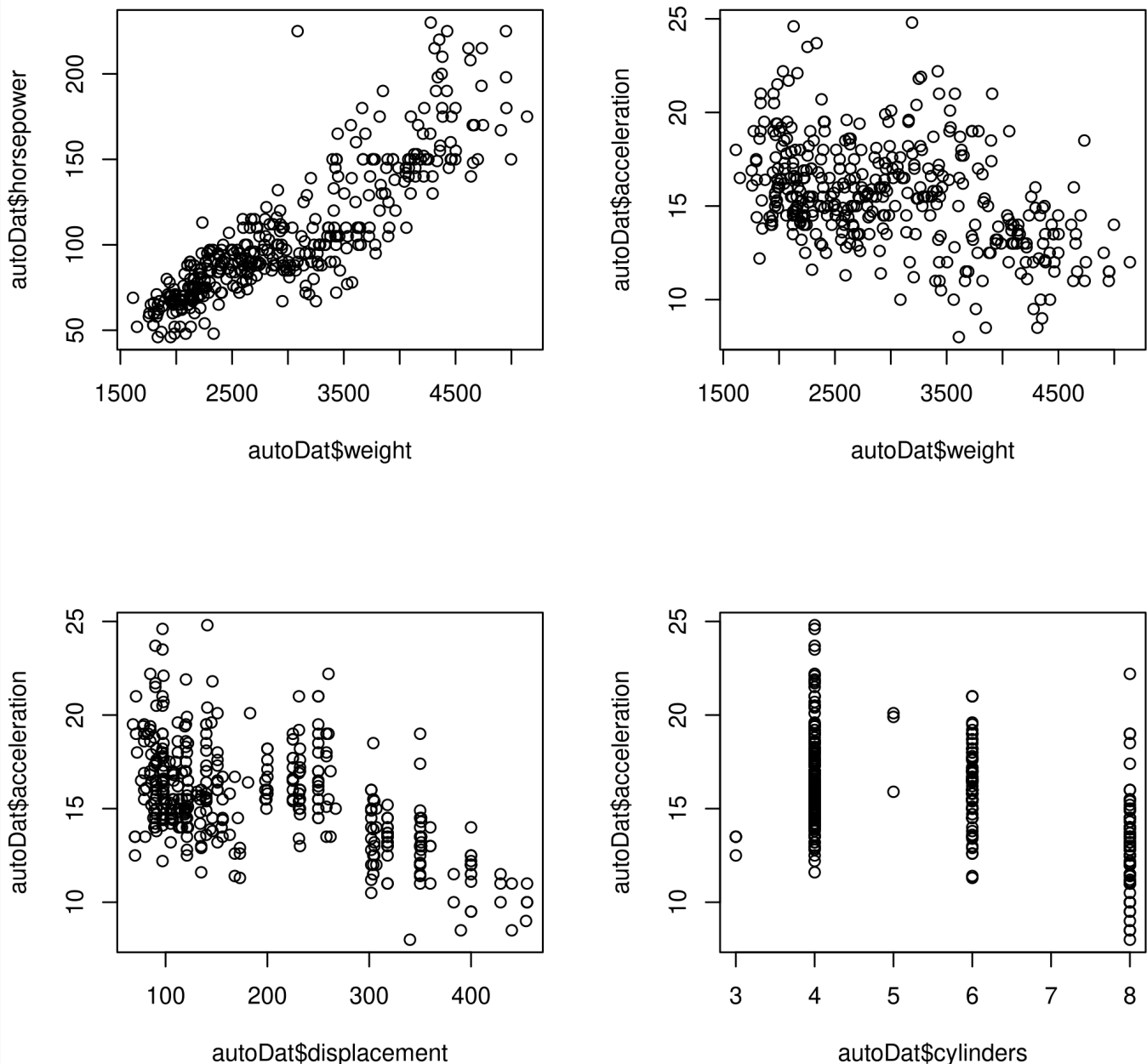


Figure 9: Relationship determination

- Evidently horsepower is directly proportional to weight but acceleration is inversely proportional to weight
- Acceleration is also inversely proportional to displacement
- Cylinders are a poor measure, not surprising since there are only 5 values

(f) Choosing predictors for gas mileage mpg

Let us recall certain key elements of the quantitative aspects of the dataset.

```
summary(noFactors) # To understand the spread
```

```
##      mpg      cylinders  displacement  horsepower      weight
## Min.   : 9.00    Min.   :3.000    Min.   : 68.0    Min.   : 46.0    Min.   :1613
## 1st Qu.:17.00    1st Qu.:4.000    1st Qu.:105.0    1st Qu.: 75.0    1st Qu.:2225
## Median :22.75    Median :4.000    Median :151.0    Median : 93.5    Median :2804
## Mean   :23.45    Mean   :5.472    Mean   :194.4    Mean   :104.5    Mean   :2978
## 3rd Qu.:29.00    3rd Qu.:8.000    3rd Qu.:275.8    3rd Qu.:126.0    3rd Qu.:3615
## Max.   :46.60    Max.   :8.000    Max.   :455.0    Max.   :230.0    Max.   :5140
## acceleration      year
## Min.   : 8.00    Min.   :70.00
## 1st Qu.:13.78    1st Qu.:73.00
## Median :15.50    Median :76.00
## Mean   :15.54    Mean   :75.98
## 3rd Qu.:17.02    3rd Qu.:79.00
## Max.   :24.80    Max.   :82.00
```

```
getLengths # To get the number of unique values
```

```
##      mpg  cylinders displacement  horsepower      weight acceleration
##      127         5         81         93         346         95
##      year      origin         name
##      13         3         301
```

From this we can assert easily that the number of cylinders is not of much interest for predictions of the mileage.

```
par(mfrow=c(3,2))
plot(noFactors$mpg,noFactors$horsepower)
plot(noFactors$mpg,noFactors$weight)
plot(noFactors$mpg,noFactors$displacement)
plot(noFactors$mpg,noFactors$acceleration)
plot(noFactors$mpg,noFactors$year)
```

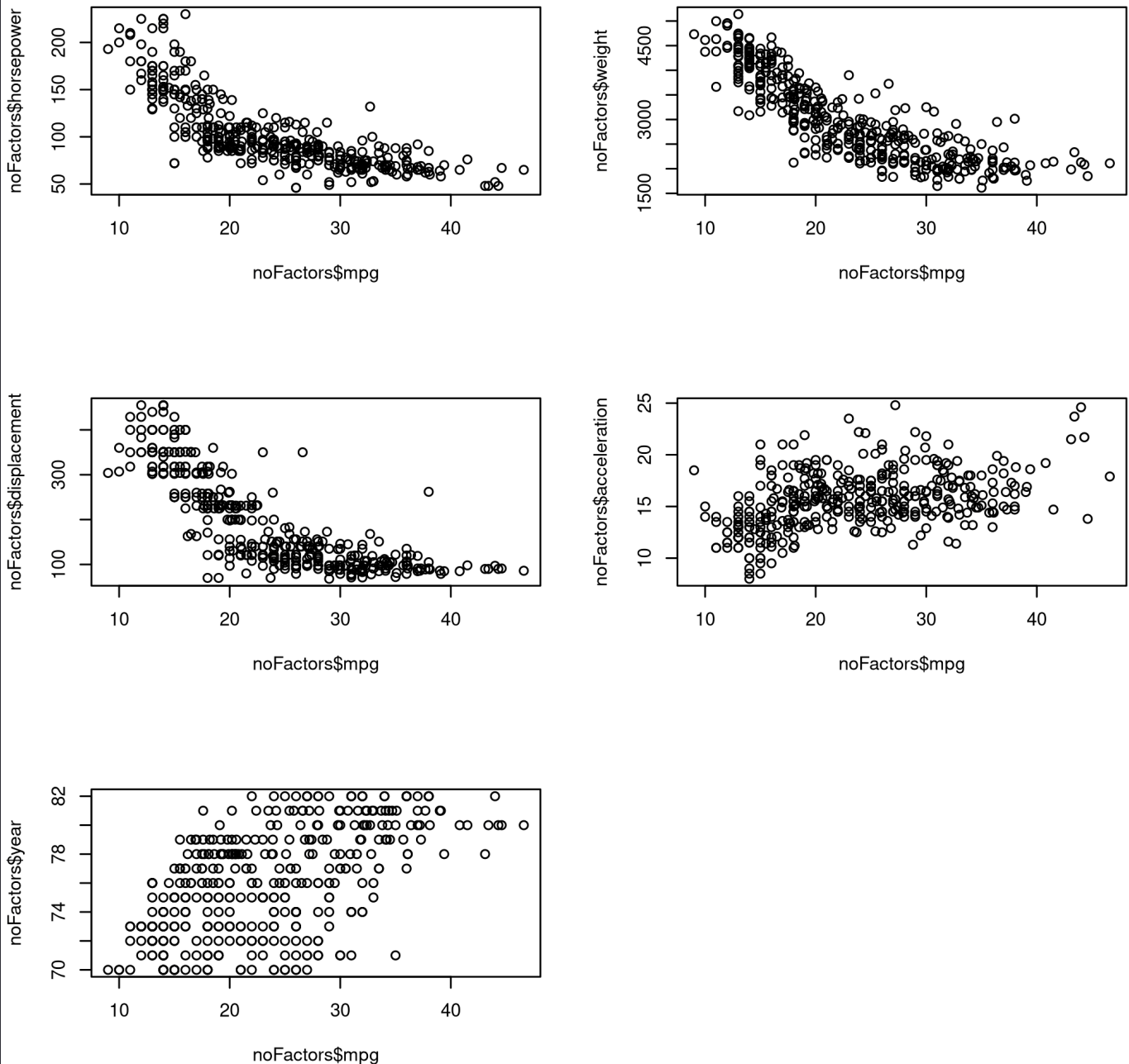



Figure 10: Predictions

- So now we know that the mileage increases when horsepower is low, weight is low, displacement is low and acceleration is high

Where low represents an inverse response and high represents a direct response.

- It is also clear that the mileage increases every year

Chapter III - Linear Regression

Question 3.9 - Page 122

This question involves the use of multiple linear regression on the Auto data set.

(a) Produce a scatterplot matrix which includes all of the variables in the data set.

(b) Compute the matrix of correlations between the variables using the function `cor()`. You will need to exclude the name variable, `cor()` which is qualitative.

(c) Use the `lm()` function to perform a multiple linear regression with `mpg` as the response and all other variables except name as the predictors. Use the `summary()` function to print the results. Comment on the output. For instance:

1. Is there a relationship between the predictors and the response?
2. Which predictors appear to have a statistically significant relationship to the response?
3. What does the coefficient for the year variable suggest?

(d) Use the `plot()` function to produce diagnostic plots of the linear regression fit. Comment on any problems you see with the fit. Do the residual plots suggest any unusually large outliers? Does the leverage plot identify any observations with unusually high leverage?

(e) Use the `*` and `:` symbols to fit linear regression models with interaction effects. Do any interactions appear to be statistically significant?

(f) Try a few different transformations of the variables, such as $\log X$, $\sqrt{X}$, X^2 . Comment on your findings.

Answer

Once again, we will use the dataset from the library.

```
cleanAuto <- na.omit(autoDat)
summary(cleanAuto) # Already created above, so no need to do na.omit again
```

```
##      mpg      cylinders  displacement  horsepower      weight
##  Min.   : 9.00    Min.   :3.000    Min.   : 68.0    Min.   : 46.0    Min.   :1613
## 1st Qu.:17.00    1st Qu.:4.000    1st Qu.:105.0    1st Qu.: 75.0    1st Qu.:2225
## Median :22.75    Median :4.000    Median :151.0    Median : 93.5    Median :2804
## Mean   :23.45    Mean   :5.472    Mean   :194.4    Mean   :104.5    Mean   :2978
## 3rd Qu.:29.00    3rd Qu.:8.000    3rd Qu.:275.8    3rd Qu.:126.0    3rd Qu.:3615
## Max.   :46.60    Max.   :8.000    Max.   :455.0    Max.   :230.0    Max.   :5140
##
##  acceleration      year      origin      name
##  Min.   : 8.00    Min.   :70.00    Min.   :1.000    amc matador      : 5
## 1st Qu.:13.78    1st Qu.:73.00    1st Qu.:1.000    ford pinto       : 5
## Median :15.50    Median :76.00    Median :1.000    toyota corolla   : 5
## Mean   :15.54    Mean   :75.98    Mean   :1.577    amc gremlin      : 4
## 3rd Qu.:17.02    3rd Qu.:79.00    3rd Qu.:2.000    amc hornet       : 4
## Max.   :24.80    Max.   :82.00    Max.   :3.000    chevrolet chevette: 4
##                                     (Other)      :365
```

(a) Scatterplot

```
pairs(cleanAuto)
```

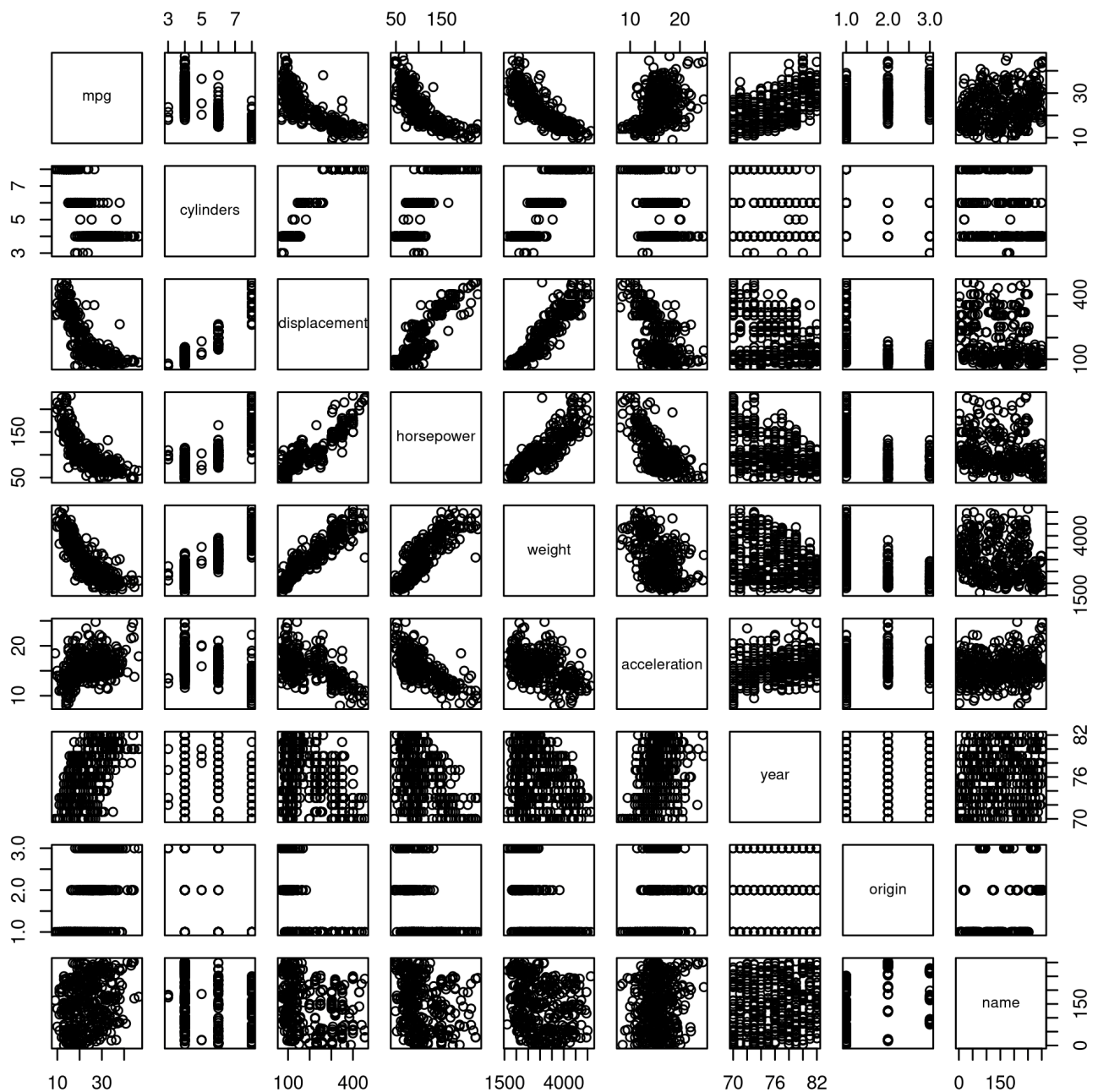


Figure 11: Scatterplot

(b) Correlation matrix. For this we exclude the qualitative variables either by using `select` or by using the existing `noFactors` dataset

```
# A full set
ISLR::Auto %>% na.omit %>% select(-name, -origin) %>% cor
```

```
##          mpg  cylinders displacement horsepower    weight
## mpg      1.0000000 -0.7776175   -0.8051269 -0.7784268 -0.8322442
## cylinders -0.7776175  1.0000000    0.9508233  0.8429834  0.8975273
## displacement -0.8051269  0.9508233    1.0000000  0.8972570  0.9329944
## horsepower -0.7784268  0.8429834    0.8972570  1.0000000  0.8645377
## weight     -0.8322442  0.8975273    0.9329944  0.8645377  1.0000000
## acceleration 0.4233285 -0.5046834   -0.5438005 -0.6891955 -0.4168392
## year        0.5805410 -0.3456474   -0.3698552 -0.4163615 -0.3091199
##          acceleration      year
## mpg          0.4233285  0.5805410
## cylinders     -0.5046834 -0.3456474
## displacement  -0.5438005 -0.3698552
## horsepower    -0.6891955 -0.4163615
## weight        -0.4168392 -0.3091199
## acceleration  1.0000000  0.2903161
## year          0.2903161  1.0000000
```

(c) Multiple Linear Regression

```
# Fit against every variable
lm.fit=lm(mpg~.,data=noFactors)
summary(lm.fit)
```

```
##
## Call:
## lm(formula = mpg ~ ., data = noFactors)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.6927 -2.3864 -0.0801  2.0291 14.3607
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.454e+01  4.764e+00  -3.051  0.00244 **
## cylinders    -3.299e-01  3.321e-01  -0.993  0.32122
## displacement  7.678e-03  7.358e-03   1.044  0.29733
## horsepower   -3.914e-04  1.384e-02  -0.028  0.97745
## weight       -6.795e-03  6.700e-04 -10.141 < 2e-16 ***
## acceleration  8.527e-02  1.020e-01   0.836  0.40383
## year         7.534e-01  5.262e-02  14.318 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.435 on 385 degrees of freedom
## Multiple R-squared:  0.8093, Adjusted R-squared:  0.8063
## F-statistic: 272.2 on 6 and 385 DF, p-value: < 2.2e-16
```

```
# Fit against one variable
noFactors %>% lm(mpg~horsepower,data=.) %>% summary
```

```
##
## Call:
## lm(formula = mpg ~ horsepower, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5710  -3.2592  -0.3435   2.7630  16.9240
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 39.935861   0.717499   55.66  <2e-16 ***
## horsepower  -0.157845   0.006446  -24.49  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.906 on 390 degrees of freedom
## Multiple R-squared:  0.6059, Adjusted R-squared:  0.6049
## F-statistic: 599.7 on 1 and 390 DF, p-value: < 2.2e-16
```

```
noFactors %>% lm(mpg~year,data=.) %>% summary
```

```
##
## Call:
## lm(formula = mpg ~ year, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.0212  -5.4411  -0.4412   4.9739  18.2088
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -70.01167    6.64516  -10.54  <2e-16 ***
## year         1.23004     0.08736   14.08  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.363 on 390 degrees of freedom
## Multiple R-squared:  0.337, Adjusted R-squared:  0.3353
## F-statistic: 198.3 on 1 and 390 DF, p-value: < 2.2e-16
```

```
noFactors %>% lm(mpg~acceleration,data=.) %>% summary
```

```
##
## Call:
## lm(formula = mpg ~ acceleration, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.989  -5.616  -1.199   4.801  23.239
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.8332     2.0485   2.359  0.0188 *
## acceleration  1.1976     0.1298   9.228  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.08 on 390 degrees of freedom
## Multiple R-squared:  0.1792, Adjusted R-squared:  0.1771
## F-statistic: 85.15 on 1 and 390 DF, p-value: < 2.2e-16
```

```
noFactors %>% lm(mpg~weight,data=.) %>% summary
```

```
##
## Call:
## lm(formula = mpg ~ weight, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.9736  -2.7556  -0.3358   2.1379  16.5194
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  46.216524   0.798673   57.87  <2e-16 ***
## weight      -0.007647   0.000258  -29.64  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.333 on 390 degrees of freedom
## Multiple R-squared:  0.6926, Adjusted R-squared:  0.6918
## F-statistic: 878.8 on 1 and 390 DF, p-value: < 2.2e-16
```

```
noFactors %>% lm(mpg~displacement,data=.) %>% summary
```

```
##
## Call:
## lm(formula = mpg ~ displacement, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.9170  -3.0243  -0.5021   2.3512  18.6128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  35.12064    0.49443   71.03  <2e-16 ***
## displacement -0.06005    0.00224  -26.81  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.635 on 390 degrees of freedom
## Multiple R-squared:  0.6482, Adjusted R-squared:  0.6473
## F-statistic: 718.7 on 1 and 390 DF, p-value: < 2.2e-16
```

1. Clearly there is a relationship between the predictors and variables, mostly as described previously, with the following broad trends:
 - Inversely proportional to Horsepower, Weight, and Displacement
2. The predictors which have a relationship to the response are (based on R squared values):

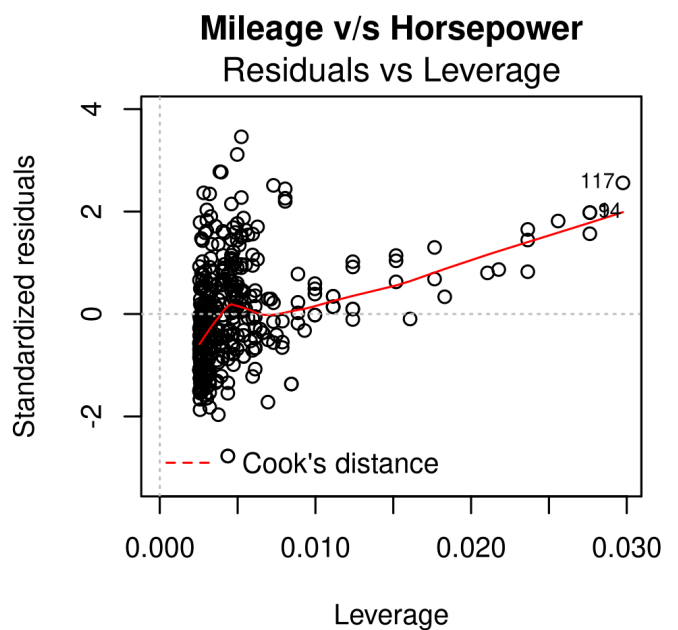
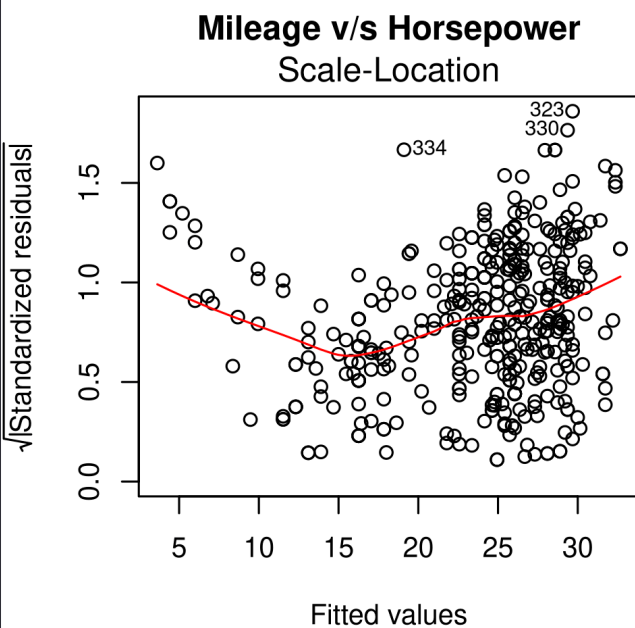
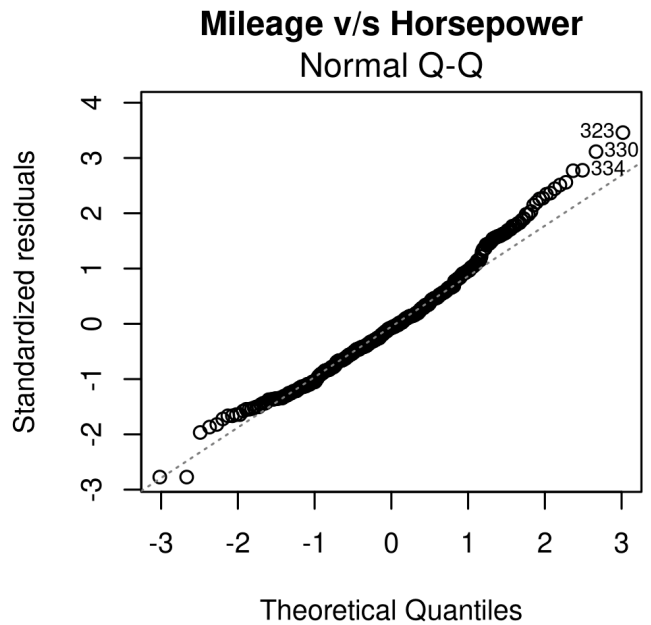
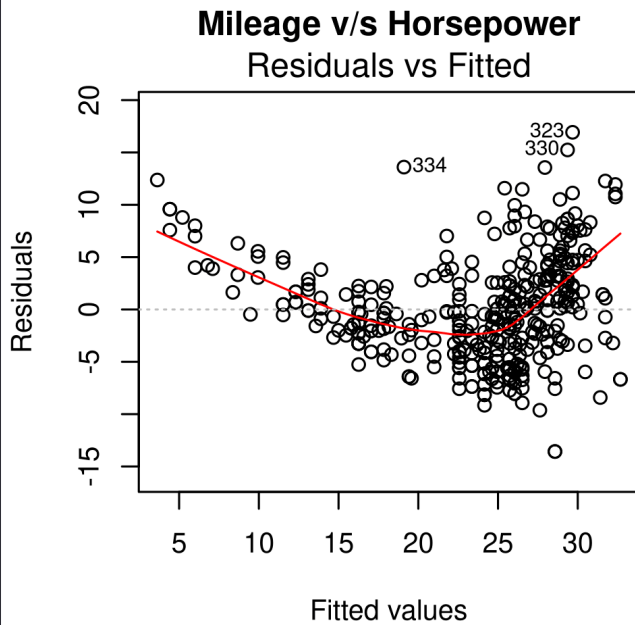
all > weight > displacement > horsepower > year > acceleration

However, things lower than horsepower are not statistically significant.

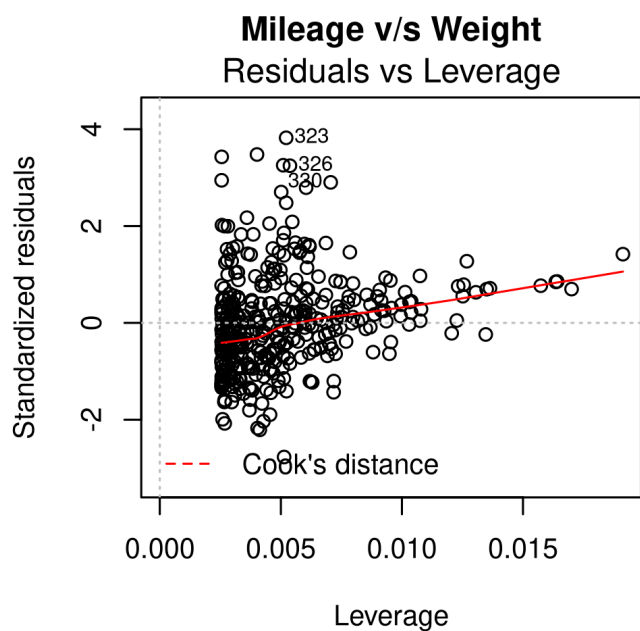
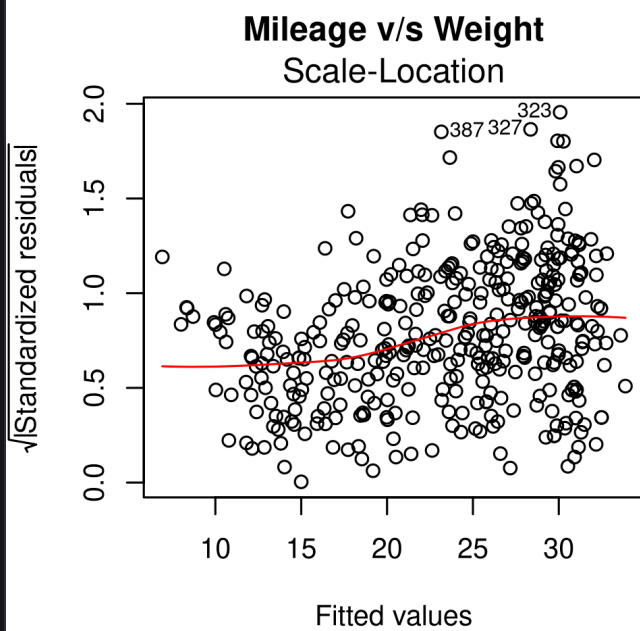
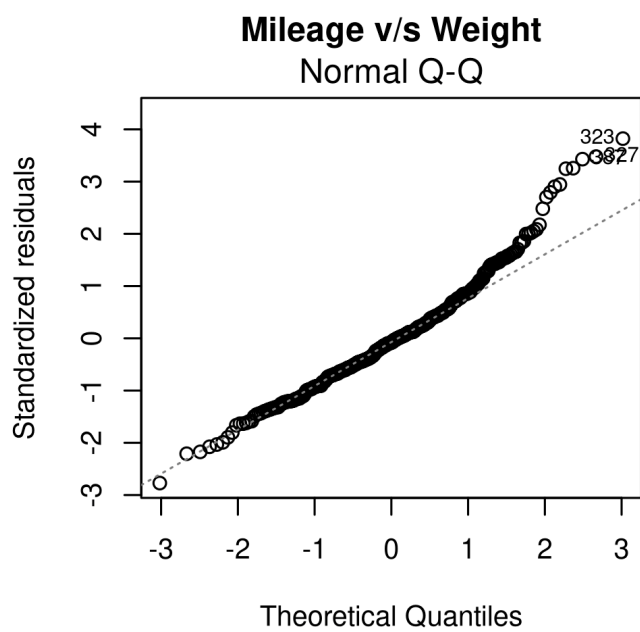
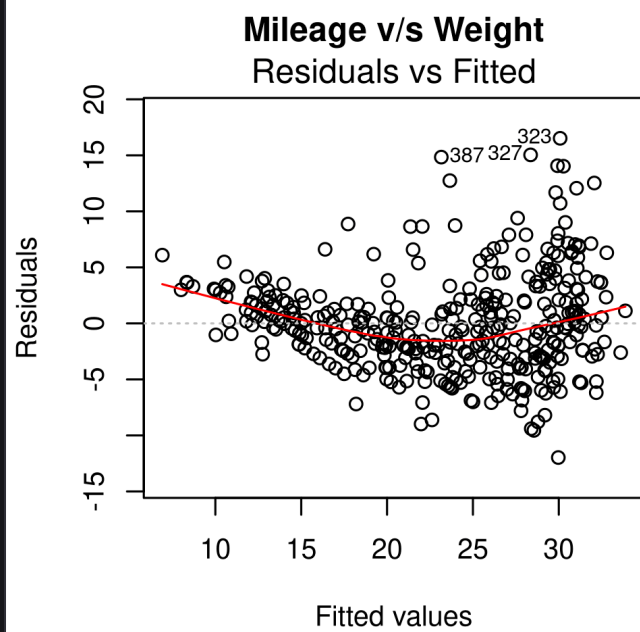
3. The visual analysis of the `year` variable suggests that the mileage grows every year. However, it is clear from the summary, that there is no statistical significance of year when used to fit a single parameter linear model. We note that when we compare this to the multiple linear regression analysis, we see that the year factor accounts for **0.7508** of the total, that is, the cars become more efficient every year

(d) Lets plot these

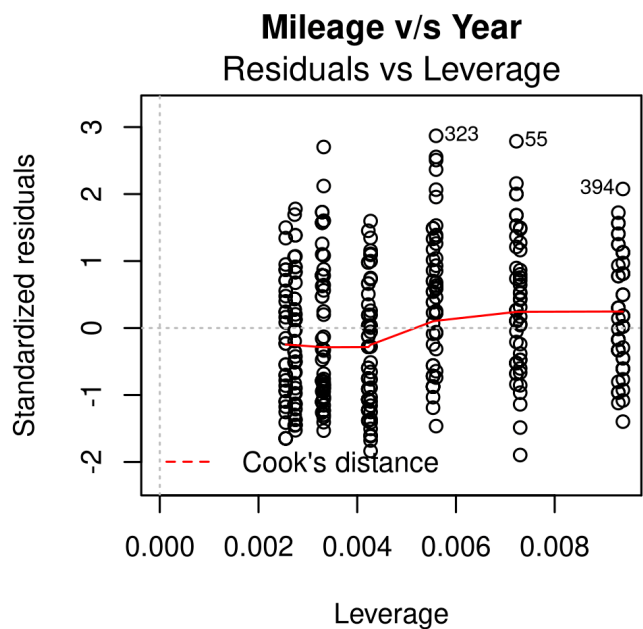
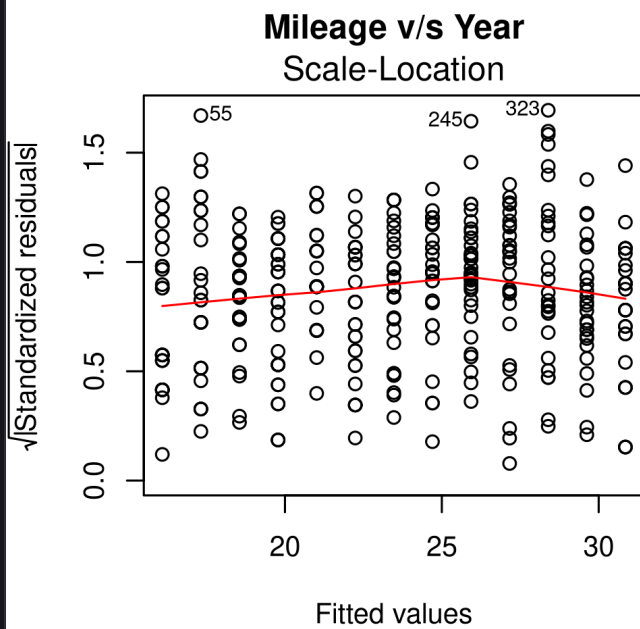
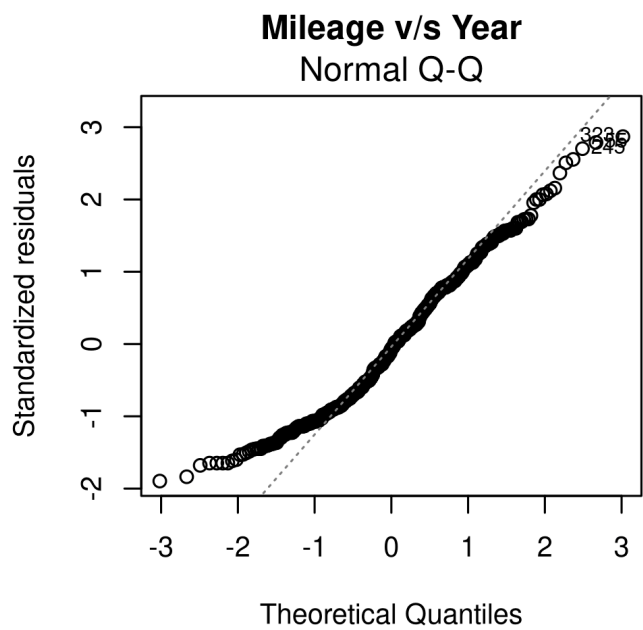
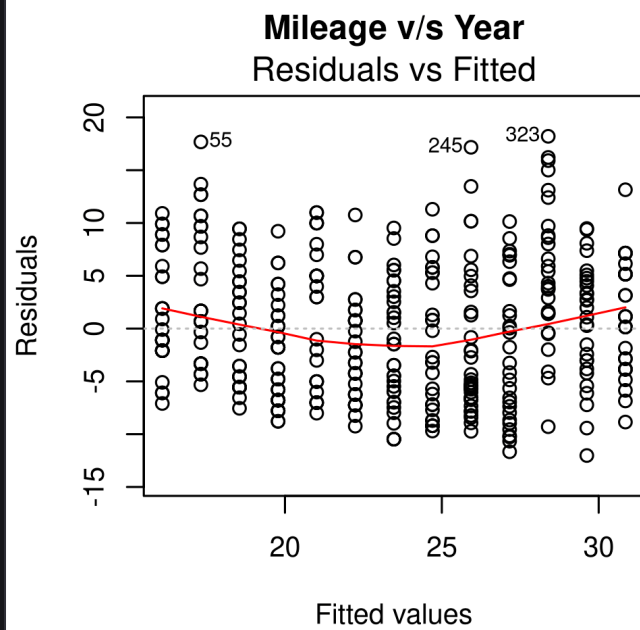
```
par(mfrow=c(2,2))
noFactors %>% lm(mpg~horsepower,data=.) %>% plot(main="Mileage v/s Horsepower")
```



```
noFactors %>% lm(mpg~weight,data=.) %>% plot(main="Mileage v/s Weight")
```



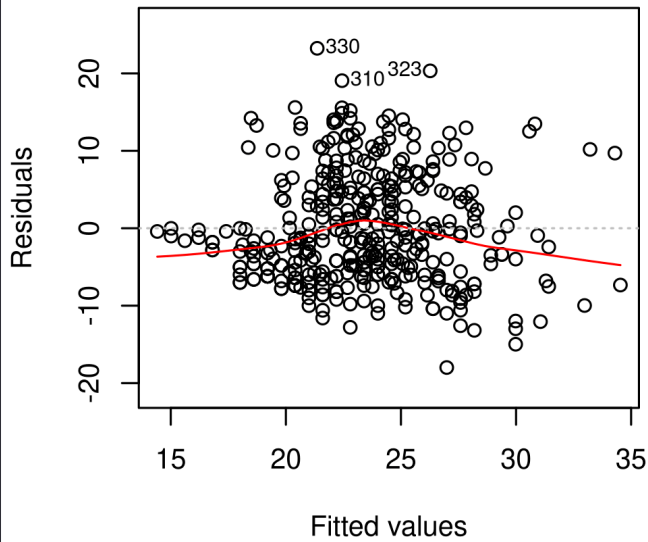
```
noFactors %>% lm(mpg~year,data=.) %>% plot(main="Mileage v/s Year")
```

```
noFactors %>% lm(mpg~acceleration,data=.) %>% plot(main="Mileage v/s Acceleration")
```

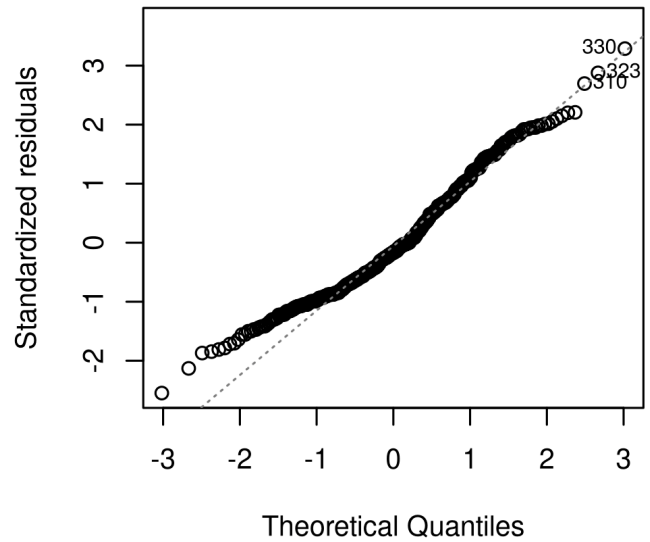
Mileage v/s Acceleration

Residuals vs Fitted



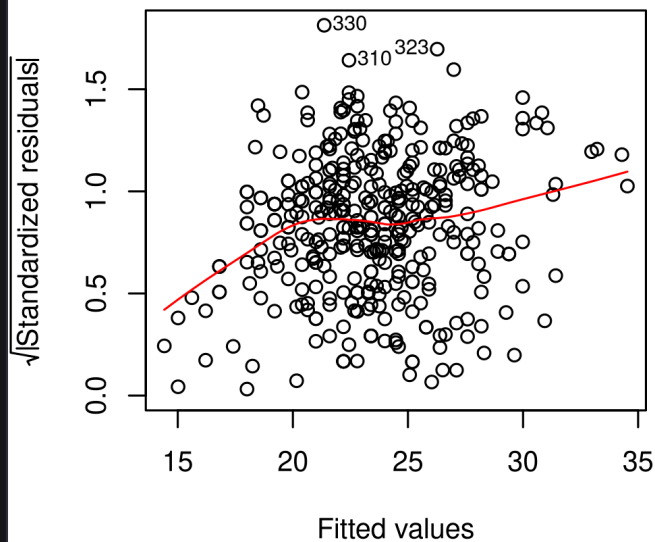
Mileage v/s Acceleration

Normal Q-Q



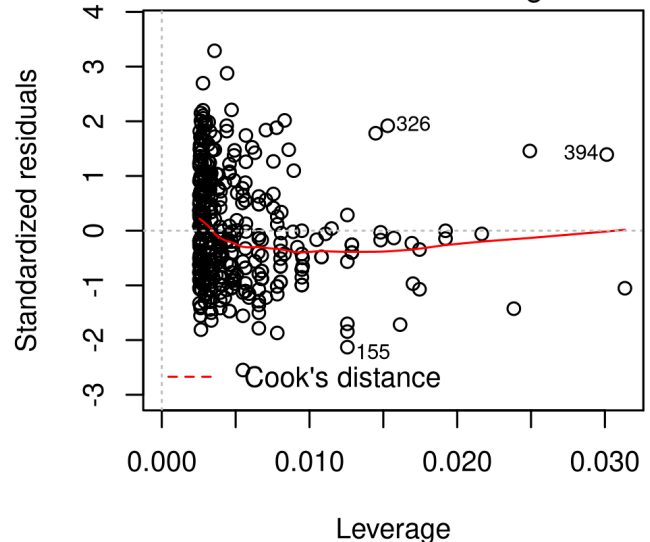
Mileage v/s Acceleration

Scale-Location



Mileage v/s Acceleration

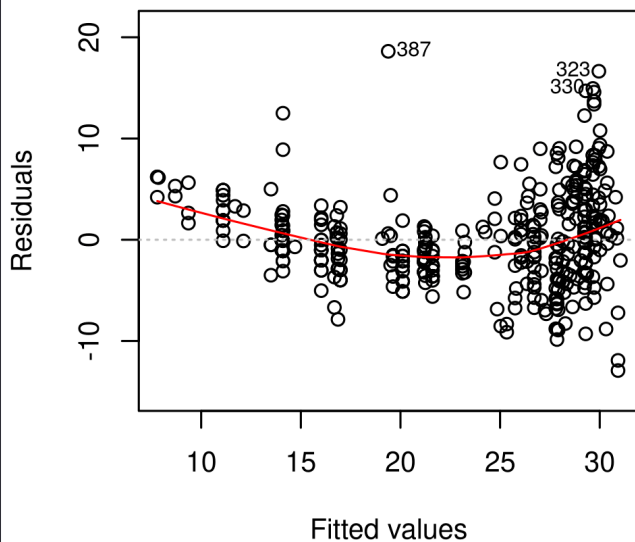
Residuals vs Leverage



```
noFactors %>% lm(mpg~displacement,data=.) %>% plot(main="Mileage v/s Displacement")
```

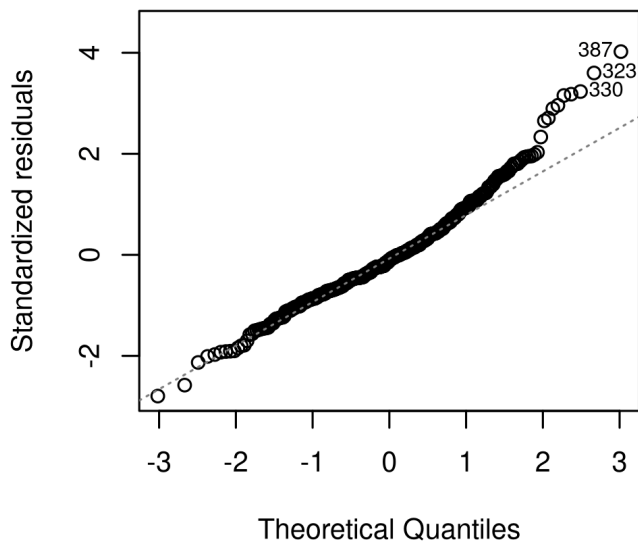
Mileage v/s Displacement

Residuals vs Fitted



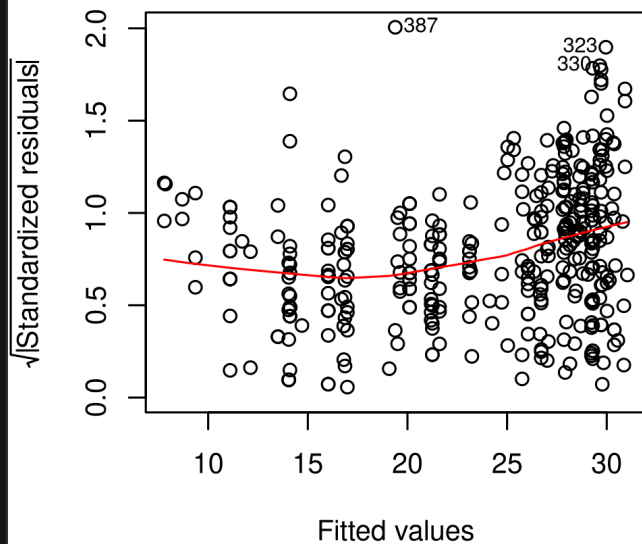
Mileage v/s Displacement

Normal Q-Q



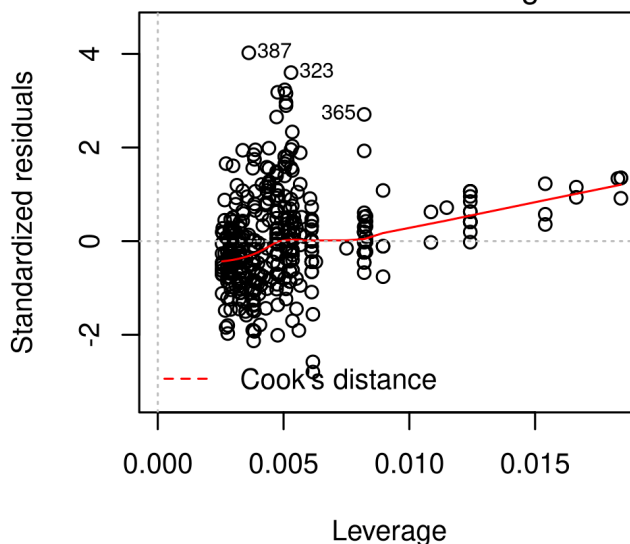
Mileage v/s Displacement

Scale-Location

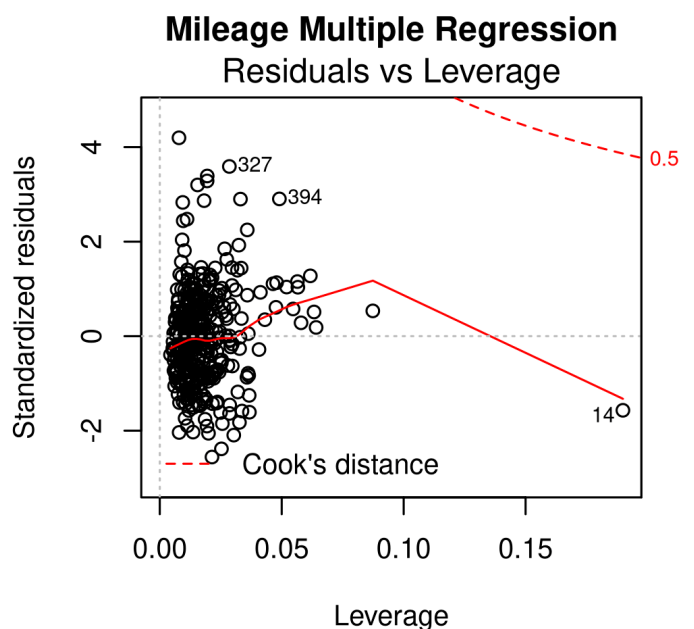
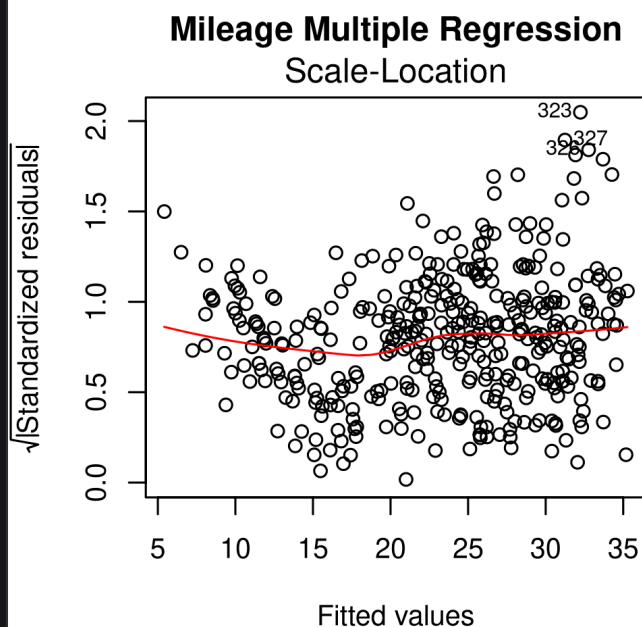
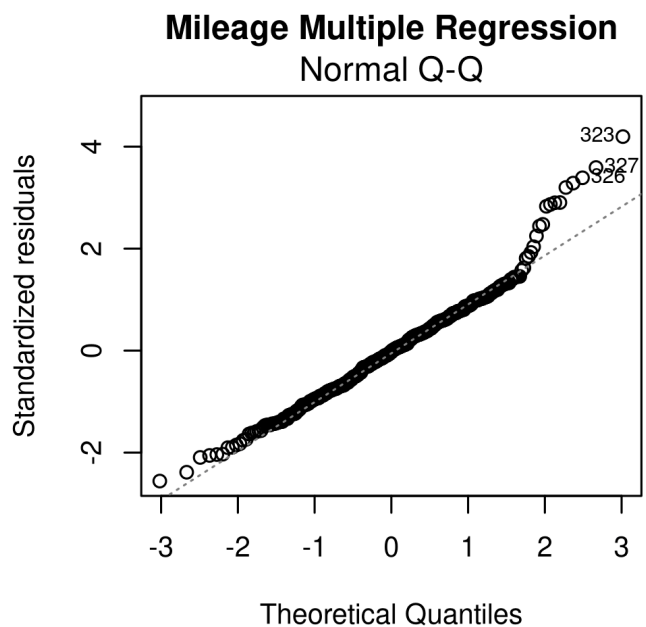
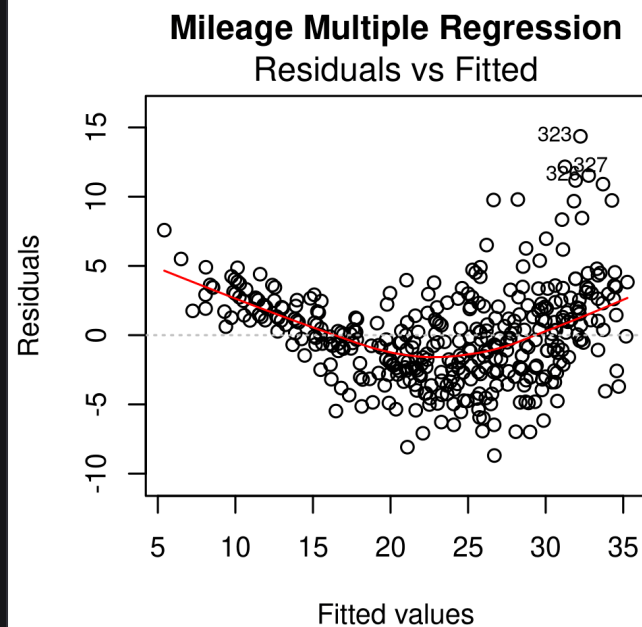


Mileage v/s Displacement

Residuals vs Leverage



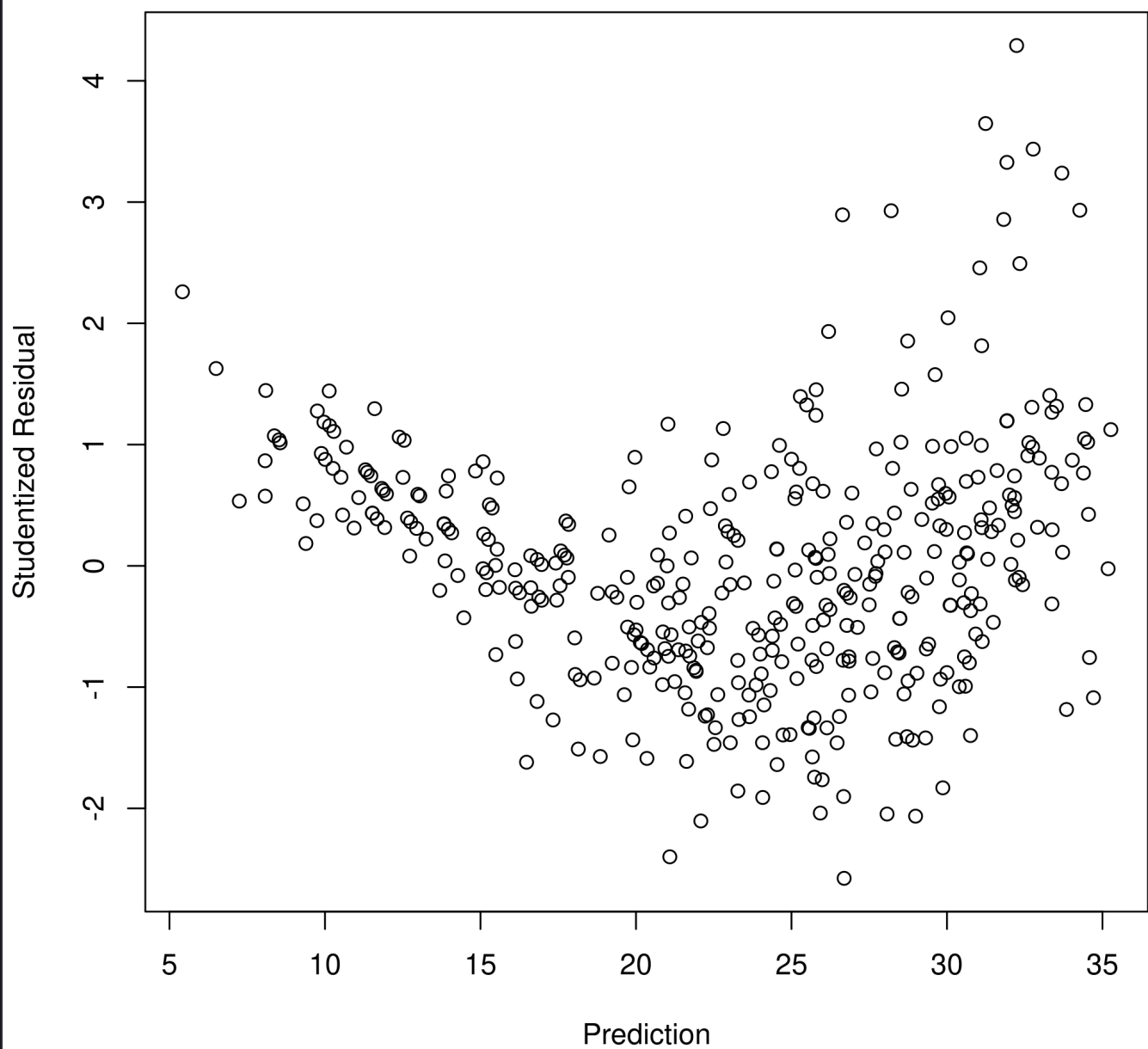
```
noFactors %>% lm(mpg~.,data=.) %>% plot(main="Mileage Multiple Regression")
```



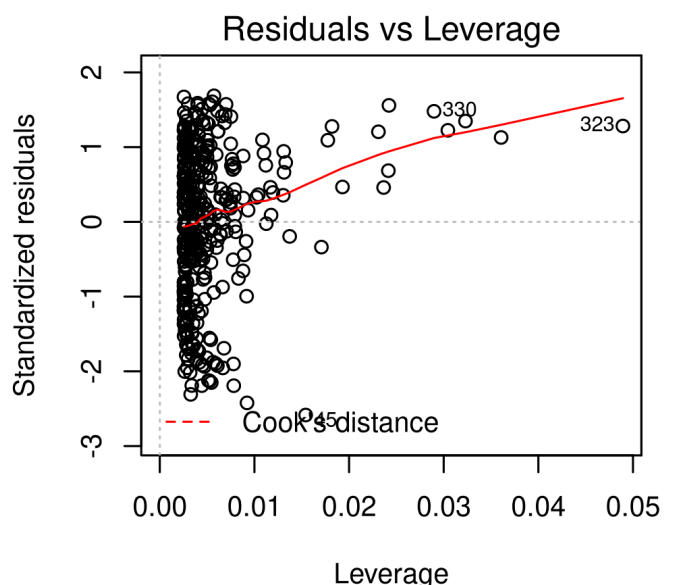
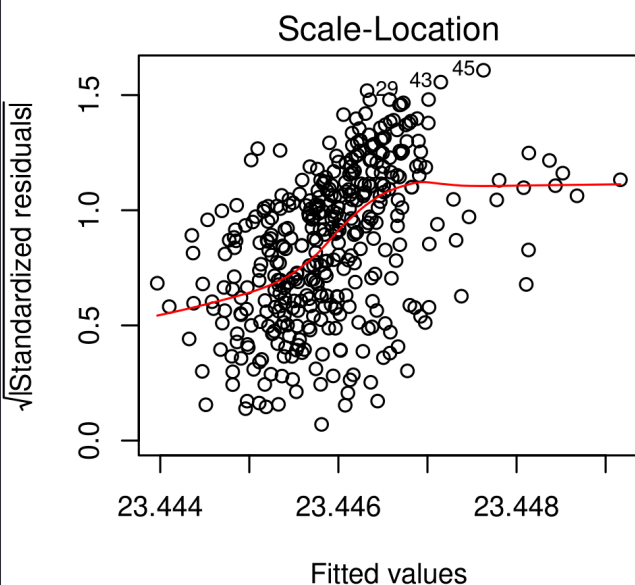
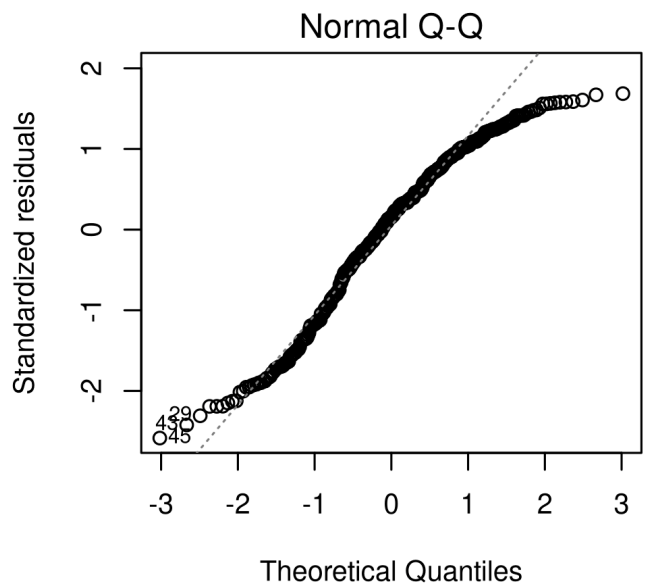
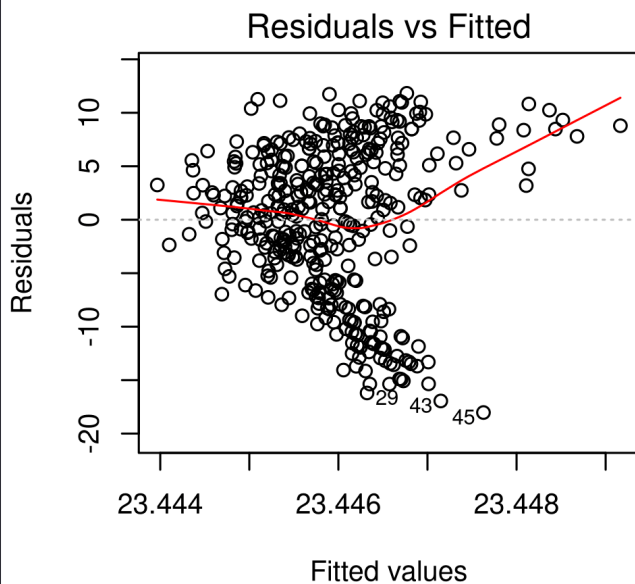
Form this we can see that the fit is not very accurate as there is a clear curve to the residuals. The 14th point has high leverage, though it is of a small magnitude. Thus it is not expected to have affected the plot too much.

We know that an observation with a studentized residual greater than 3 in absolute value are possible outliers. Hence we must plot this.

```
# Predict and get the plot
fitPlot <- noFactors %>% lm(mpg~.,data=.)
# See residuals
plot(xlab="Prediction",ylab="Studentized Residual",x=predict(fitPlot),y=rstudent(fitPlot))
```



```
# Try a linear fit of studentized residuals
par(mfrow=c(2,2))
plot(lm(predict(fitPlot)~rstudent(fitPlot)))
```



Clearly the studentized residuals are nonlinear w.r.t the prediction. Also, some points are above the absolute value of 3 so they might be outliers, in keeping with the leverage plot.

(e) Interaction effects

We recall that $x*y$ corresponds to $x+y+x:y$

```
# View the correlation matrix
cleanAuto %>% select(-name,-origin) %>% cor
```

```
##          mpg  cylinders displacement horsepower    weight
## mpg      1.0000000 -0.7776175   -0.8051269 -0.7784268 -0.8322442
## cylinders -0.7776175  1.0000000    0.9508233  0.8429834  0.8975273
## displacement -0.8051269  0.9508233    1.0000000  0.8972570  0.9329944
## horsepower -0.7784268  0.8429834    0.8972570  1.0000000  0.8645377
## weight     -0.8322442  0.8975273    0.9329944  0.8645377  1.0000000
## acceleration 0.4233285 -0.5046834   -0.5438005 -0.6891955 -0.4168392
## year        0.5805410 -0.3456474   -0.3698552 -0.4163615 -0.3091199
##          acceleration    year
## mpg          0.4233285  0.5805410
## cylinders    -0.5046834 -0.3456474
## displacement -0.5438005 -0.3698552
## horsepower   -0.6891955 -0.4163615
## weight       -0.4168392 -0.3091199
## acceleration  1.0000000  0.2903161
## year          0.2903161  1.0000000
```

```
summary(lm(mpg~weight*displacement*year,data=noFactors[(10:85),]))
```

```
##
## Call:
## lm(formula = mpg ~ weight * displacement * year, data = noFactors[(10:85),
##    ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3020 -0.9055  0.0966  0.8912  3.7049
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.961e+02  2.578e+02   1.537   0.129
## weight         -1.030e-01  1.008e-01  -1.021   0.311
## displacement   -1.587e+00  1.308e+00  -1.213   0.229
## year           -4.889e+00  3.623e+00  -1.349   0.182
## weight:displacement  3.926e-04  3.734e-04   1.051   0.297
## weight:year        1.317e-03  1.418e-03   0.929   0.356
## displacement:year   2.150e-02  1.846e-02   1.165   0.248
## weight:displacement:year -5.287e-06  5.253e-06  -1.007   0.318
##
## Residual standard error: 1.8 on 68 degrees of freedom
## Multiple R-squared:  0.922, Adjusted R-squared:  0.914
## F-statistic: 114.9 on 7 and 68 DF, p-value: < 2.2e-16
```

```
summary(lm(mpg~weight*displacement*year,data=noFactors))
```

```
##
## Call:
## lm(formula = mpg ~ weight * displacement * year, data = noFactors)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6093 -1.6472 -0.0531  1.2289 14.5604
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -8.437e+01  3.128e+01  -2.697   0.0073 **
## weight         8.489e-03  1.322e-02   0.642   0.5212
## displacement  3.434e-01  1.969e-01   1.744   0.0820 .
## year          1.828e+00  4.127e-01   4.430 1.23e-05 ***
## weight:displacement -6.589e-05  5.055e-05  -1.303   0.1932
## weight:year      -2.433e-04  1.744e-04  -1.395   0.1638
## displacement:year  -5.566e-03  2.674e-03  -2.082   0.0380 *
## weight:displacement:year 1.144e-06  6.823e-07   1.677   0.0944 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.951 on 384 degrees of freedom
## Multiple R-squared:  0.8596, Adjusted R-squared:  0.8571
## F-statistic: 336 on 7 and 384 DF, p-value: < 2.2e-16
```

- Adding the interaction effects of the **3** most positive R value terms improves the existing prediction to be better than that obtained by considering all effects.
- We note that the best model is obtained by removing the range identified in chapter 2.

(f) Nonlinear transformations

```
summary(lm(mpg~weight*displacement*year+I(year^2),data=noFactors[(10:85),]))
```

```
##
## Call:
## lm(formula = mpg ~ weight * displacement * year + I(year^2),
##      data = noFactors[(10:85), ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.1815 -0.8235  0.0144  1.0076  3.9420
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -4.205e+03  1.810e+03  -2.324   0.0232 *
## weight         -8.800e-02  9.709e-02  -0.906   0.3680
## displacement  -1.030e+00  1.276e+00  -0.807   0.4225
## year           1.238e+02  5.026e+01   2.464   0.0163 *
## I(year^2)      -9.000e-01  3.506e-01  -2.567   0.0125 *
## weight:displacement  2.471e-04  3.634e-04   0.680   0.4988
## weight:year        1.113e-03  1.365e-03   0.815   0.4177
## displacement:year   1.368e-02  1.800e-02   0.760   0.4501
## weight:displacement:year -3.254e-06  5.111e-06  -0.637   0.5264
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.73 on 67 degrees of freedom
## Multiple R-squared:  0.929, Adjusted R-squared:  0.9205
## F-statistic: 109.6 on 8 and 67 DF, p-value: < 2.2e-16
```



```
summary(lm(mpg~.-I(log(acceleration^2)),data=noFactors[(10:85),]))
```

```
##
## Call:
## lm(formula = mpg ~ . - I(log(acceleration^2)), data = noFactors[(10:85),
##    ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.232 -1.470 -0.211  1.075  7.088
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.3787633  24.1208720   1.715   0.0907 .
## cylinders      0.0863161   0.6112822   0.141   0.8881
## displacement -0.0148491   0.0103249  -1.438   0.1549
## horsepower   -0.0158500   0.0151259  -1.048   0.2984
## weight       -0.0039125   0.0008546  -4.578 2.02e-05 ***
## acceleration -0.1473786   0.1438220  -1.025   0.3091
## year         -0.0378187   0.3380266  -0.112   0.9112
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.262 on 69 degrees of freedom
## Multiple R-squared:  0.8751, Adjusted R-squared:  0.8642
## F-statistic: 80.55 on 6 and 69 DF,  p-value: < 2.2e-16
```

- The best model I found was still the one without the non-linear transformation but with removed outliers and additional interaction effects of `displacement`, `year` and `weight`
- A popular approach is to use a `log` transform for both the inputs and the outputs

```
summary(lm(log(mpg)~.,data=noFactors[(10:85),]))
```

```
##
## Call:
## lm(formula = log(mpg) ~ ., data = noFactors[(10:85), ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.285805 -0.052358 -0.001456  0.066521  0.209739
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.886e+00  1.028e+00   3.781 0.000328 ***
## cylinders     -1.771e-02  2.604e-02  -0.680 0.498669
## displacement -1.540e-04  4.399e-04  -0.350 0.727314
## horsepower    -2.343e-03  6.444e-04  -3.636 0.000529 ***
## weight        -1.960e-04  3.641e-05  -5.383 9.51e-07 ***
## acceleration -1.525e-02  6.128e-03  -2.489 0.015224 *
## year          4.138e-03  1.440e-02   0.287 0.774703
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09636 on 69 degrees of freedom
## Multiple R-squared:  0.919, Adjusted R-squared:  0.912
## F-statistic: 130.5 on 6 and 69 DF,  p-value: < 2.2e-16
```

```
summary(lm(log(mpg)~log(weight*displacement*year),data=noFactors[(10:85),]))
```

```
##
## Call:
## lm(formula = log(mpg) ~ log(weight * displacement * year), data = noFactors[(10:85),
##      ])
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -0.41121 -0.04107  0.01266  0.07791  0.21056
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   8.91995    0.26467   33.70  <2e-16 ***
## log(weight * displacement * year) -0.34250    0.01508  -22.71  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1158 on 74 degrees of freedom
## Multiple R-squared:  0.8745, Adjusted R-squared:  0.8728
## F-statistic: 515.6 on 1 and 74 DF,  p-value: < 2.2e-16
```

Question 3.10 - Page 123

This question should be answered using the Carseats data set.

- (a) Fit a multiple regression model to predict `Sales` using `Price`, `Urban`, and `US`.
- (b) Provide an interpretation of each coefficient in the model. Be careful—some of the variables in the model are qualitative!
- (c) Write out the model in equation form, being careful to handle the qualitative variables properly.
- (d) For which of the predictors can you reject the null hypothesis $H_0 : \beta_j = 0$?
- (e) On the basis of your response to the previous question, fit a smaller model that only uses the predictors for which there is evidence of association with the outcome.
- (f) How well do the models in (a) and (e) fit the data?
- (g) Using the model from (e), obtain 95 confidence intervals for the coefficient(s).
- (h) Is there evidence of outliers or high leverage observations in the model from (e)?

Answer

Load the dataset (and clean it)

```
cleanCarSeats <- na.omit(ISLR::Carseats)
```

Obtain summary statistics

```
cleanCarSeats %>% sapply(unique) %>% sapply(length)
```

```
##      Sales  CompPrice      Income Advertising  Population      Price
##      336         73         98          28         275         101
## ShelfLoc      Age  Education      Urban      US
##      3         56          9          2          2
```

```
str(cleanCarSeats)
```

```
## 'data.frame':    400 obs. of  11 variables:
## $ Sales      : num  9.5 11.22 10.06 7.4 4.15 ...
## $ CompPrice  : num  138 111 113 117 141 124 115 136 132 132 ...
## $ Income     : num   73 48 35 100 64 113 105 81 110 113 ...
## $ Advertising: num   11 16 10 4 3 13 0 15 0 0 ...
## $ Population : num  276 260 269 466 340 501 45 425 108 131 ...
## $ Price      : num  120 83 80 97 128 72 108 120 124 124 ...
## $ ShelfLoc   : Factor w/ 3 levels "Bad","Good","Medium": 1 2 3 3 1 1 3 2 3 3 ...
## $ Age       : num   42 65 59 55 38 78 71 67 76 76 ...
## $ Education  : num   17 10 12 14 13 16 15 10 10 17 ...
## $ Urban      : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 1 2 2 1 1 ...
## $ US         : Factor w/ 2 levels "No","Yes": 2 2 2 2 1 2 1 2 1 2 ...
```

```
summary(cleanCarSeats)
```

```
##      Sales      CompPrice      Income      Advertising
## Min.   : 0.000   Min.   : 77   Min.   : 21.00   Min.   : 0.000
## 1st Qu.: 5.390   1st Qu.:115   1st Qu.: 42.75   1st Qu.: 0.000
## Median : 7.490   Median :125   Median : 69.00   Median : 5.000
## Mean   : 7.496   Mean   :125   Mean   : 68.66   Mean   : 6.635
## 3rd Qu.: 9.320   3rd Qu.:135   3rd Qu.: 91.00   3rd Qu.:12.000
## Max.   :16.270   Max.   :175   Max.   :120.00   Max.   :29.000
## Population      Price      ShelfLoc      Age      Education
## Min.   : 10.0   Min.   : 24.0   Bad   : 96   Min.   :25.00   Min.   :10.0
## 1st Qu.:139.0   1st Qu.:100.0   Good  : 85   1st Qu.:39.75   1st Qu.:12.0
## Median :272.0   Median :117.0   Medium:219   Median :54.50   Median :14.0
## Mean   :264.8   Mean   :115.8               Mean   :53.32   Mean   :13.9
## 3rd Qu.:398.5   3rd Qu.:131.0               3rd Qu.:66.00   3rd Qu.:16.0
## Max.   :509.0   Max.   :191.0               Max.   :80.00   Max.   :18.0
## Urban      US
## No :118   No :142
## Yes:282   Yes:258
##
##
##
##
```

We can see that:

- **Urban**, **US** and **ShelveLoc** are factors with 2,2 and 3 levels respectively
- Education has only 9 unique values so we might as well consider it to be a factor too if we need to

(a) Multiple Regression Model

Fit it to things

```
summary(lm(Sales~.,data=cleanCarSeats))
```

```
##
## Call:
## lm(formula = Sales ~ ., data = cleanCarSeats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8692 -0.6908  0.0211  0.6636  3.4115
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.6606231   0.6034487   9.380 < 2e-16 ***
## CompPrice      0.0928153   0.0041477  22.378 < 2e-16 ***
## Income         0.0158028   0.0018451   8.565 2.58e-16 ***
## Advertising    0.1230951   0.0111237  11.066 < 2e-16 ***
## Population     0.0002079   0.0003705   0.561  0.575
## Price         -0.0953579   0.0026711 -35.700 < 2e-16 ***
## ShelfLocGood   4.8501827   0.1531100  31.678 < 2e-16 ***
## ShelfLocMedium 1.9567148   0.1261056  15.516 < 2e-16 ***
## Age           -0.0460452   0.0031817 -14.472 < 2e-16 ***
## Education      -0.0211018   0.0197205  -1.070  0.285
## UrbanYes       0.1228864   0.1129761   1.088  0.277
## USYes         -0.1840928   0.1498423  -1.229  0.220
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.019 on 388 degrees of freedom
## Multiple R-squared:  0.8734, Adjusted R-squared:  0.8698
## F-statistic: 243.4 on 11 and 388 DF, p-value: < 2.2e-16
```

```
summary(lm(Sales~US*Price*Urban,data=cleanCarSeats))
```

```
##
## Call:
## lm(formula = Sales ~ US * Price * Urban, data = cleanCarSeats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.7952 -1.6659 -0.0984  1.6119  7.2433
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13.456350   1.727210   7.791 6.03e-14 ***
## USYes          2.049051   2.322591   0.882  0.378
## Price         -0.061657   0.014875  -4.145 4.17e-05 ***
## UrbanYes       -0.651545   2.071401  -0.315  0.753
## USYes:Price    -0.001567   0.019972  -0.078  0.937
## USYes:UrbanYes -1.122034   2.759662  -0.407  0.685
## Price:UrbanYes  0.010793   0.017796   0.606  0.545
## USYes:Price:UrbanYes 0.001288   0.023619   0.055  0.957
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.473 on 392 degrees of freedom
## Multiple R-squared:  0.2467, Adjusted R-squared:  0.2333
## F-statistic: 18.34 on 7 and 392 DF, p-value: < 2.2e-16
```

```
summary(lm(Sales~US+Price+Urban,data=cleanCarSeats))
```

```
##
## Call:
## lm(formula = Sales ~ US + Price + Urban, data = cleanCarSeats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.9206 -1.6220 -0.0564  1.5786  7.0581
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.043469   0.651012  20.036 < 2e-16 ***
## USYes       1.200573   0.259042   4.635 4.86e-06 ***
## Price      -0.054459   0.005242 -10.389 < 2e-16 ***
## UrbanYes    -0.021916   0.271650  -0.081  0.936
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.472 on 396 degrees of freedom
## Multiple R-squared:  0.2393, Adjusted R-squared:  0.2335
## F-statistic: 41.52 on 3 and 396 DF, p-value: < 2.2e-16
```

(b) Interpret stuff

To interpret the data, we need to determine which of the models fits the data best, we will use `anova()` to test this:

```
lmCarSAll<-lm(Sales~.,data=cleanCarSeats)
lmCarStimesPUU<-lm(Sales~US*Price*Urban,data=cleanCarSeats)
lmCarSplusPUU<-lm(Sales~US+Price+Urban,data=cleanCarSeats)
anova(lmCarSAll,lmCarStimesPUU,lmCarSplusPUU)
```

```
## Analysis of Variance Table
##
## Model 1: Sales ~ CompPrice + Income + Advertising + Population + Price +
##      ShelveLoc + Age + Education + Urban + US
## Model 2: Sales ~ US * Price * Urban
## Model 3: Sales ~ US + Price + Urban
##      Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      388  402.83
## 2      392 2397.10 -4   -1994.27 480.2082 < 2.2e-16 ***
## 3      396 2420.83 -4     -23.73   5.7149 0.0001772 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(lmCarStimesPUU,lmCarSplusPUU)
```

```
## Analysis of Variance Table
##
## Model 1: Sales ~ US * Price * Urban
## Model 2: Sales ~ US + Price + Urban
##      Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      392 2397.1
## 2      396 2420.8 -4     -23.734 0.9703 0.4236
```

Remember that it is not possible to use `anova()` unless the same variables are present in all the models being tested, so it is meaningless to use `anova` for `lmCarSAll` along with the others, because we can't change the interaction model to get only the main effects.

- We note that due to the low value of the F-statistic and the non-zero value of the p-value we cannot disregard the null hypothesis, or in other words, the models are basically the same in terms of their performance.

This means that I would like to continue with the simpler model, since the increase in R squared is too small to account for dealing with the additional factors.

- We see immediately, that there is a positive correlation only with being in the US
- Increases in price and being in an urban area actually decrease the sales, which is not surprising since being in the an urban area is probably correlated to a higher price, which we can check immediately

```
summary(lm(Price~Urban,data=cleanCarSeats))
```

```
##
## Call:
## lm(formula = Price ~ Urban, data = cleanCarSeats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -92.514 -15.514   1.205  14.595  74.486
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  114.076      2.180  52.330  <2e-16 ***
## UrbanYes      2.438      2.596   0.939   0.348
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.68 on 398 degrees of freedom
## Multiple R-squared:  0.002211, Adjusted R-squared:  -0.0002965
## F-statistic: 0.8817 on 1 and 398 DF, p-value: 0.3483
```

We see that our assumption is validated. Being in an urban area has a low t-statistic for a positive increase on the slope

- Returning to our previous model, we note that there is a high value of the p-value of the t-statistic for `Urban` being true, this means there isn't a real relationship between being in an urban area and the sales. This makes intuitive sense as well

note `t-test` is essentially a linear model with one variable, that is, if we want to find out if there is a relation between having a store in an urban area, we could sum all the urban yes and divide by the number of observations and compare that to the sum of all the urban no divided by the number of observations which is essentially the `t-test` again.

- `Price` is significant, and has an inverse relation with the sales, so we should keep that in mind

(c) In Equation Form:

$$Sales = 1.200573 \backslash *USYes - 0.054459 \backslash *Price - 0.021916 \backslash *UrbanYes + 13.043469$$

(e) Other models

- We know from our case-study on testing the full multiple linear regression for `Sales` that there are definitely more important variables being ignored. However, we also know that `Urban` is not significant, so we can use a smaller model.

```
lmCarSplusPU<-lm(Sales~US+Price, data=cleanCarSeats)
```

(f) Comparison of models

```
summary(lmCarSplusPU)
```

```
##
## Call:
## lm(formula = Sales ~ US + Price, data = cleanCarSeats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.9269 -1.6286 -0.0574  1.5766  7.0515
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.03079    0.63098  20.652  < 2e-16 ***
## USYes       1.19964    0.25846   4.641 4.71e-06 ***
## Price      -0.05448    0.00523 -10.416  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.469 on 397 degrees of freedom
## Multiple R-squared:  0.2393, Adjusted R-squared:  0.2354
## F-statistic: 62.43 on 2 and 397 DF,  p-value: < 2.2e-16
```

```
anova(lmCarSplusPUU,lmCarSplusPU)
```

```
## Analysis of Variance Table
##
## Model 1: Sales ~ US + Price + Urban
## Model 2: Sales ~ US + Price
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     396 2420.8
## 2     397 2420.9 -1   -0.03979 0.0065 0.9357
```

As expected, the low value of the F statistic and the high p-value for the `anova()` test asserts that the null hypothesis cannot be neglected, thus there are no differences between the model with the insignificant parameter, which is also seen in the `R` squared value, which is the same for both models

(g) Confidence Intervals

```
confint(lmCarSplusPU)
```

```
##              2.5 %      97.5 %
## (Intercept) 11.79032020 14.27126531
## USYes       0.69151957  1.70776632
## Price      -0.06475984 -0.04419543
```

```
confint(lmCarSplusPUU)
```

```
##                2.5 %      97.5 %
## (Intercept) 11.76359670 14.32334118
## USYes       0.69130419  1.70984121
## Price      -0.06476419 -0.04415351
## UrbanYes   -0.55597316  0.51214085
```

- ☐ Look into trying to plot this with `ggplot`

(h) Outliers

- We will first check the leverage plots

```
par(mfrow=c(2,2))
plot(lmCarSplusPU)
```

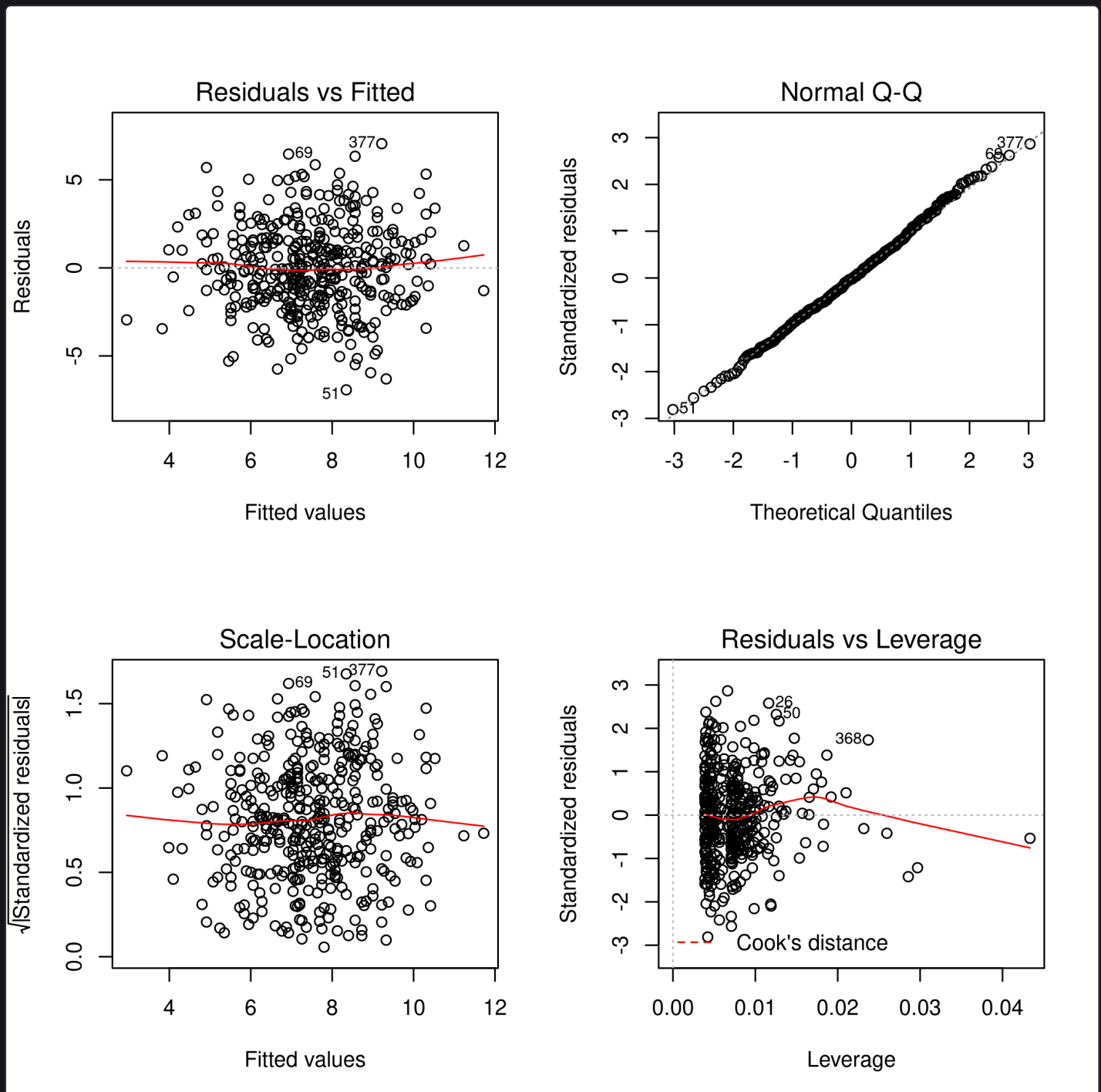


Figure 12: Leverage Plots

We can see there is a point with high leverage, but it has a low residual. In any case we should check further.

- Now we will check the studentized residuals to see if they are greater than 3

```
# See residuals  
plot(xlab="Prediction",ylab="Studentized Residual",x=predict(lmCarSplusPU),y=rstudent(lmCarSplusPU))
```

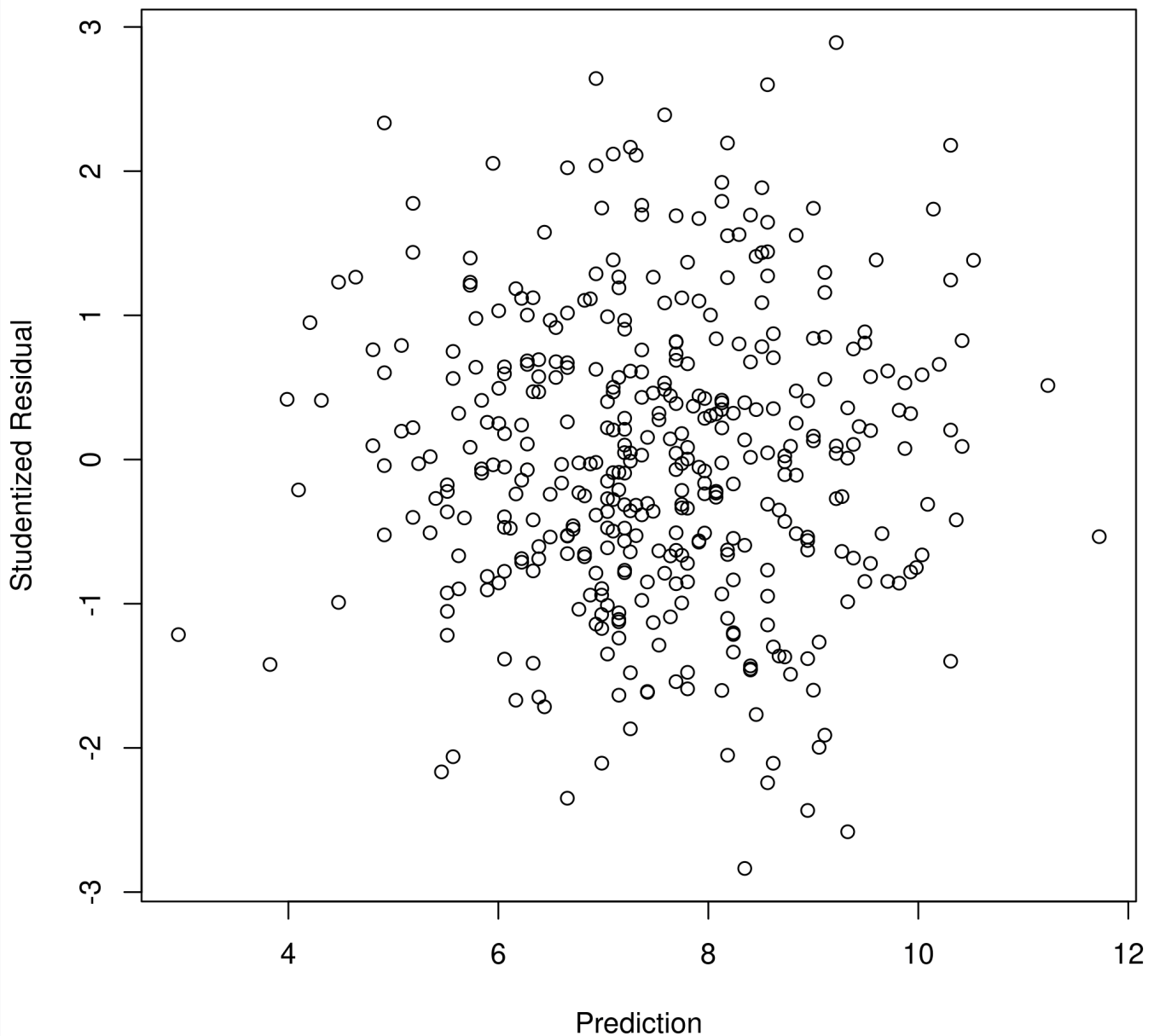


Figure 13: Studentized residuals

Thus I would say there are no outliers in our dataset, as none of our datapoints have an absolute studentized residual above 3.